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Optimal distribution estimates for commutators and Marcinkiewicz multipliers

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Abstract
The main objective of this paper is to investigate the distributional estimates for (i) commutators with Calderón–Zygmund integral operators; (ii) Marcinkiewicz multipliers; (iii) Littlewood–Paley square function, via semigroup $\{\mathscr{C}^\alpha\}_{\alpha>0}$ generated by Cesàro operator. In each of the cases (i)–(iii) we obtain new estimates of the distribution of elements in the range of the underlying operators in terms of the distribution function of the input function. Our method allows us to obtain optimal estimates shedding additional light at the results due to Pérez (J Funct Anal 128(1):163–185, 1995), Tao and Wright (Rev Mat Iberoam 17(3):521–558, 2001), Bakas et al. (Rev Mat Iberoam 2024), Bourgain (On the Behavior of the Constant in the Littlewood–Paley Inequality, Geometric Aspects of Functional Analysis (1987–88), Lecture Notes in Math., vol. 1376, pp. 202–208. Springer, Berlin, 1989). The main feature of the distributional form inequalities lies in its broad applicability across diverse problems in analysis, e.g. they allow obtaining estimates in wide range of symmetric quasi-Banach interpolation spaces between L_p and L_q $(1 < p < q < \infty)$, not just for L_p-spaces $(1 < p < \infty)$.

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1 Introduction

1.1 Historical background and main results

The main objective of this paper is to investigate the distributional estimates related to three distinct operators, namely, the commutators with Calderón–Zygmund integral operators, Marcinkiewicz multipliers, and Littlewood–Paley square function. The discussion in terms of distributional estimates of the elements in the range of a given operator in terms of the distribution of its preimage has a rich history. Indeed, it is well-known that, for each locally integrable function $f \in (L_1 + L_\infty)(\mathbb{R}^d)$ and the Hardy–Littlewood maximal function M on $\mathbb{R}^d$, there are constants $c_d^{(1)}$ and $c_d^{(2)}$ depending on d such that

$$c_d^{(1)} \mathscr{C}\mu(f) \le \mu(Mf) \le c_d^{(2)} \mathscr{C}\mu(f), \tag{1.1}$$

where $\mu(f)$ stands for the right continuous decreasing rearrangement of the function $|f|$ ([6, 31, 32] and Sect. 2 below), and $\mathscr{C}$ is the Cesàro operator given by

$$\mathscr{C}g(t) = \frac{1}{t}\int_0^t g(s)ds, \quad g \in (L_1 + L_\infty)(0,\infty).$$

The right-hand side of (1.1) goes back to Hardy, Littlewood and Riesz [21, 39], and the left-hand side of (1.1) is due to Herz [23] (see also [5] for a simple proof). Numerous estimates of type (1.1) for other classical operators have been studied; we refer the reader to Boyd [9] and Bennett–Rudnick [4] for similar estimates of Hilbert transform, and Riesz transform to O'Neil–Weiss [36] and Bennett–Sharpley [5], see also [6, 14, 35] for other analogous results. Recently, the optimality estimates of type (1.1) and related applications for Haar martingale transforms were established in [1], and, in [18, 43], one can find more applications of such optimal estimates.

In the present paper, we firstly concern with the optimal estimates of type (1.1) for a commutator with a Calderón–Zygmund integral operator. Recall that a standard Calderón–Zygmund integral operator T is an integral operator given by

$$Tf(x) = \int_{\mathbb{R}^d} K(x,y)f(y)dy, \quad x \notin \operatorname{supp}(f),$$

which is bounded on $L_2(\mathbb{R}^d)$ and such that the kernel function $K : \mathbb{R}^d \times \mathbb{R}^d \setminus \{(x,x) : x \in \mathbb{R}^d\} \to \mathbb{C}$ satisfies the following conditions

Assumption 1.1 There is a constant $c_T > 0$ such that

(i) for $x \neq y$ in $\mathbb{R}^d$, $|K(x,y)| \le \frac{c_T}{|x-y|^d}$;
(ii) for $|x-y| \ge 2|x-x'| > 0$,

$$\big|K(x,y) - K(x',y)\big| + \big|K(y,x) - K(y,x')\big| \le c_T \frac{|x-x'|}{|x-y|^{d+1}}.$$

A standard Calderón–Zygmund integral operator T is of strong type (p, p), $1 < p < \infty$, and of weak type $(1, 1)$ (see e.g. [11, 13] or [20, Theorem 8.2.1]).

Theorem A [11, 13] *Let T be a standard Calderón–Zygmund integral operator with kernel K satisfying Assumption 1.1. There exists a positive constant c_T such that*

$$\|T\|_{L_1(\mathbb{R}^d)\to L_{1,\infty}(\mathbb{R}^d)} \le c_T$$

and

$$\|T\|_{L_p(\mathbb{R}^d)\to L_p(\mathbb{R}^d)} \le c_T \max\{p', p\}, \quad 1 < p < \infty.$$

Combining Theorem A and certain extrapolation tricks (see for instance [43, Theorem 14]), one can obtain the following distributional estimate:

$$\mu(Tf) \le c_T(\mathscr{C} + \mathscr{C}^*)\mu(f), \quad \mu(f) \in \mathrm{dom}(\mathscr{C}) \cap \mathrm{dom}(\mathscr{C}^*). \tag{1.2}$$

Here $\mathscr{C}$ denotes the Cesàro operator and $\mathscr{C}^*$ denotes its formal dual (see Sect. 2.4 below). A weaker result than (1.2) was obtained by Calderón [10], the exact estimate (1.2) can be found in [4, 41] (see also [30, 40]). The estimate (1.2) is optimal in the following sense that (see e.g. [6, p. 140] or [41]): if $T = H$ is the Hilbert transform and $f \in L_0(\mathbb{R})$, the space of measurable functions on $\mathbb{R}$, such that $\mu(f) \in \mathrm{dom}(\mathscr{C}) \cap \mathrm{dom}(\mathscr{C}^*)$, then there exists a function $g \in L_0(\mathbb{R})$ such that $\mu(g) = \mu(f)$ and

$$(\mathscr{C} + \mathscr{C}^*)\mu(f) \le 2\pi\mu(Hg), \tag{1.3}$$

where the Hilbert transform H is given by the formula

$$(Hf)(x) = \text{p.v.}\frac{1}{\pi}\int_{\mathbb{R}} \frac{f(y)}{x - y}dy.$$

Let $T : L_2(\mathbb{R}^d) \to L_2(\mathbb{R}^d)$ be a standard Calderón–Zygmund integral operator. For a locally integrable function ϕ on $\mathbb{R}^d$, the commutator $[M_\phi, T]$ is given by

$$[M_\phi, T] = M_\phi T - TM_\phi$$

where M_ϕ is the multiplication operator by ϕ on $L_2(\mathbb{R}^d)$. In the foundational paper [15], Coifman, Rochberg and Weiss showed that, for Riesz transform $T = R_j := \partial_j \Delta^{-\frac{1}{2}}$, $[M_\phi, T]$ is bounded on $L_p(\mathbb{R}^d)$, $1 < p < \infty$, if and only if ϕ is a $\mathrm{BMO}(\mathbb{R}^d)$ function (see Sect. 2 for details). However, as shown in Pérez's familiar paper [37, Section 5], the commutator $[M_\phi, T]$ usually fails to be of weak type $(1, 1)$ even if $\phi \in \mathrm{BMO}(\mathbb{R}^d)$, but satisfies the following $L\log^+ L(\mathbb{R}^d)$ estimate

Theorem B [37, Theorem 1.1] *Let T be a standard Calderón–Zygmund integral operator with kernel K satisfying Assumption 1.1. There exists a positive constant c_T such*

that

$$m\left(\left\{y \in \mathbb{R}^d : \left|[M_\phi, T]f\right|(y) > \lambda\right\}\right) \leq c_T \int_{\mathbb{R}^d} \frac{\|\phi\|_{\mathrm{BMO}}|f(y)|}{\lambda}\left(1 + \log^+\left(\frac{\|\phi\|_{\mathrm{BMO}}|f(y)|}{\lambda}\right)\right) dy$$

for every $f \in L_1^{\mathrm{loc}}(\mathbb{R}^d)$, *for every* $\phi \in \mathrm{BMO}(\mathbb{R}^d)$ *and for every* $\lambda > 0$.

Our first main result in this paper recasts Theorem B in an unexpected form and exposes its deep connection with the behaviour of the operator $\mathscr{C}^2 + (\mathscr{C}^*)^2$. In fact, the distribution of the function $[M_\phi, T]f$ is controlled by that of $(\mathscr{C}^2 + (\mathscr{C}^*)^2)\mu(f)$. This should be compared with the equivalences (1.1), (1.2) and (1.3).

Theorem 1.2 *Let* T *be a standard Calderón–Zygmund integral operator with kernel* K *satisfying Assumption* 1.1, *and let* $\phi \in \mathrm{BMO}(\mathbb{R}^d)$. *There exists a positive constant* c_T *such that*

$$\mu([M_\phi, T]f) \leq c_T\|\phi\|_{\mathrm{BMO}} \cdot \left(\mathscr{C}^2 + (\mathscr{C}^*)^2\right)\mu(f)$$

for every $f \in L_0(\mathbb{R}^d)$ *with* $\mu(f) \in \mathrm{dom}(\mathscr{C}^2) \cap \mathrm{dom}((\mathscr{C}^*)^2)$. *This inequality is optimal: let* $d = 1$, $T = H$ *and define* $\phi \in \mathrm{BMO}(\mathbb{R})$ *by setting* $\phi(x) = \log(|x|)$, $x \in \mathbb{R}$. *For every* $f = \mu(f) \in \mathrm{dom}(\mathscr{C}^2) \cap \mathrm{dom}((\mathscr{C}^*)^2)$, *we have*

$$\mu([M_\phi, H]f) \geq \frac{1}{\pi}(\mathscr{C}^2 + (\mathscr{C}^*)^2)\mu(f).$$

Remark 1.3 We draw reader's attention to the fact that our formulation of Theorem B (due to Perez) rectifies an error in the original statement of [37, Theorem 1.1]. Theorem B can be deduced from Theorem 1.2.

The second theme of this paper is concerned with the distribution of the elements from the Schwartz function class $\mathcal{S}(\mathbb{R})$ on $\mathbb{R}$ under the action of Marcinkiewicz multipliers. Let $\mathsf{m} : \mathbb{R} \to \mathbb{C}$ be a bounded function on $\mathbb{R}$, and let T_{m} be the associated multiplier:

$$\widehat{T_{\mathsf{m}}(f)} = \mathsf{m}\widehat{f}, \quad f \in \mathcal{S}(\mathbb{R}).$$

We say T_{m} is a Marcinkiewicz multiplier whenever $m : \mathbb{R} \to \mathbb{C}$ has bounded variation on each Littlewood–Paley interval $L_k = (-2^{k+1}, -2^k] \cup [2^k, 2^{k+1})$, uniformly in $k \in \mathbb{Z}$. It is well-known that Marcinkiewicz multiplier T_{m} is bounded on $L_p(\mathbb{R})$ for $1 < p < \infty$; see e.g. [19, Theorem 8.3.1]. However, this multiplier does not map $L_1(\mathbb{R})$ to $L_{1,\infty}(\mathbb{R})$. As an appropriate substitute, in [44, Theorem 1.6], Tao and Wright proved that a Marcinkiewicz multiplier T_{m} maps $L\log^{\frac{1}{2}} L(K)$ to $L_{1,\infty}(K)$ for each compact set K of $\mathbb{R}$. Recently, Bakas et al. [3] managed to extend this result to the spaces $L\log^{\frac{1}{2}} L(\mathbb{R})$ and $L_{1,\infty}(\mathbb{R})$ as follows

Theorem C [3, 44] *Let T_{m} be a Marcinkiewicz multiplier. There is a positive constant $c_{T_{\mathsf{m}}}$ relying on* m *such that*

$$m\left(\{y \in \mathbb{R} : \ |T_{\mathsf{m}}(f)(y)| > \lambda\}\right) \leq c_{T_{\mathsf{m}}} \int_{\mathbb{R}} \frac{|f(x)|}{\lambda} \log^{\frac{1}{2}} \Big(e + \frac{|f(x)|}{\lambda}\Big) dx$$

for every $f \in L\log^{\frac{1}{2}} L(\mathbb{R})$ and for every $\lambda > 0$.

Our second main result Theorem 1.4 is a substantial improvement of Theorem C. The fractional power of Cesàro operator $\mathscr{C}^{\alpha}$ and its dual are defined in Sect. 2.4 below (see (2.4) and (2.5) there).

Theorem 1.4 *Let T_{m} be a Marcinkiewicz multiplier on $\mathbb{R}$. There is a positive constant $c_{T_{\mathsf{m}}}$ relying on* m *such that*

$$\mu(T_{\mathsf{m}} f) \leq c_{T_{\mathsf{m}}} (\mathscr{C}^{\frac{3}{2}} + (\mathscr{C}^*)^{\frac{3}{2}}) \mu(f)$$

for every $f \in L_0(\mathbb{R}^d)$ with $\mu(f) \in \mathrm{dom}(\mathscr{C}^{\frac{3}{2}}) \cap \mathrm{dom}((\mathscr{C}^*)^{\frac{3}{2}})$.

Our third objective in this paper is to specialize our techniques used in the proofs of Theorems 1.2 and 1.4 to a new approach to Bourgain's result concerning the classical Littlewood–Paley square function $S(f)$, which is defined by the formula

$$S(f) = \Big(\sum_{k \in \mathbb{Z}} |\chi_{L_k}(D) f|^2 \Big)^{\frac{1}{2}}, \quad f \in \mathcal{S}(\mathbb{R}), \tag{1.4}$$

where $L_k = (-2^{k+1}, -2^k] \cup [2^k, 2^{k+1})$ is the Littlewood–Paley interval, and $\chi_{L_k}(D)$ denotes the spectral measure of the differential operator D on $\mathbb{R}$. The L_p-boundedness of S is well known, and we recall the following remarkable result from [8], which provides the optimal norm estimates of the operator S acting on $L_p(\mathbb{R})$, $1 < p < \infty$.

Theorem D [8] *Let S be as in (1.4). We have*

$$\|S\|_{L_p(\mathbb{R}) \to L_p(\mathbb{R})} \sim \frac{1}{(p-1)^{\frac{3}{2}}}, \quad 1 < p < 2$$

and

$$\|S\|_{L_p(\mathbb{R}) \to L_p(\mathbb{R})} \sim p, \quad 2 < p < \infty$$

Our third (and last) main result yields the distribution of the square function Sf whenever $f \in L_1^{\mathrm{loc}}(\mathbb{R})$. It is an improvement of Bourgain's theorem mentioned above. Recall that there is deep connection between Marcinkiewicz multiplier theory and Littlewood–Paley square function theory (see for instance [19, 42]). However, comparing Theorems 1.4 and 1.5, we see that $T_{\mathsf{m}} f$ and $S(f)$ have different distribution estimates. We explain the underlying reason in Remark 6.10.

Theorem 1.5 *Let S be as in (1.4). There exists an absolute constant $c_{abs} > 0$ such that*

$$\mu(Sf) \le c_{abs}(\mathscr{C}^{\frac{3}{2}} + \mathscr{C}^*)\mu(f)$$

for every measurable function f on $\mathbb{R}$ with $\mu(f) \in \mathrm{dom}(\mathscr{C}^{\frac{3}{2}}) \cap \mathrm{dom}(\mathscr{C}^*)$.

Taking into account the boundedness of $\mathscr{C}^\alpha$ and $(\mathscr{C}^*)^\beta$ (see (2.6) and (2.8) below), we have the following direct corollary of Theorems 1.2, 1.4 and 1.5. Recall that a quasi-Banach space E is an interpolation space (see e.g. [31]) between L_p and L_q $(1 \le p < q \le \infty)$ if $L_p \cap L_q \subset E \subset L_p + L_q$ and for every linear operator $T : L_p + L_q \to L_p + L_q$, the boundedness of $T : L_p \to L_p$ and $T : L_q \to L_q$ implies the boundedness of $T : E \to E$.

Corollary 1.6 *Let E be a symmetric quasi-Banach function space which is an interpolation space between L_p and L_q for some $1 < p < q < \infty$. Let T, T_{m} and S be the Calderón–Zygmund integral operator, Marcinkiewicz multiplier and Littlewood–Paley square function operator as in Theorems 1.2, 1.4 and 1.5, respectively. Operators $[M_\phi, T]$ with $\phi \in \mathrm{BMO}(\mathbb{R}^d)$, T_{m} and S are all bounded on E.*

1.2 Technical innovation

Our main technical innovation underlying the proofs of Theorems 1.2, 1.4 and 1.5 is the identifying of the following notion, which serves as a bridge between Theorems A, B, and C.

Definition 1.7 Let $(\Omega, \mathcal{F}, \nu)$ be a measure space. We say that a linear operator $\mathcal{Z}$ defined on the space $L_0(\Omega, \mathcal{F}, \nu)$ of all measurable functions on $(\Omega, \mathcal{F}, \nu)$ satisfies the weak Cesàro property with parameter $\alpha > 0$ (denoted by $\mathrm{WCP}(\alpha)$) if for every $f \in L_0(\Omega, \mathcal{F}, \nu)$ supported on a set of finite measure and for every $\lambda \ge \mathscr{C}^\alpha \mu(f)(\nu(\mathrm{supp}(f)))$, we have

$$\lambda\nu\left(\{x \in \Omega : |\mathcal{Z}f|(x) > \lambda\}\right) \le \nu(\mathrm{supp}(f)) \cdot \mathscr{C}^\alpha \mu(f)(\nu(\mathrm{supp}(f))). \tag{1.5}$$

For $\alpha = 1$, the operator $\mathcal{Z}$ has the property $\mathrm{WCP}(1)$ iff it is of weak type $(1, 1)$. Since Calderón–Zygmund integral operator T is of weak type $(1, 1)$, it follows that T satisfies $\mathrm{WCP}(1)$. In Theorem 5.1 below, we show that the commutator $[M_\phi, T]$ possesses $\mathrm{WCP}(2)$. In Proposition 6.6, we prove that each Marcinkiewicz multiplier T_{m} satisfies $\mathrm{WCP}(\frac{3}{2})$. The discovery of weak Cesàro property led us to an abstract extrapolation result, i.e. Theorem 3.1, which is the crucial tool in the proofs of Theorems 1.2 and 1.4. In Lemma 5.7, we demonstrate that weak Cesàro property is the right tool of endpoint estimates and subsequent applications in the interpolation theory.

1.3 Outline of the paper and notation

Section 2 contains necessary background (like John–Nirenberg inequality and dyadic Calderón–Zygmund decomposition). We introduce fractional Cesàro operator in Sect. 2.4.

In Sect. 3, using weak Cesàro property, we establish an abstract extrapolation result Theorem 3.1, which is one of the key ingredients in the proofs of Theorems 1.2, 1.4 and 1.5.

In Sect. 4, we present an estimate associated with some Orlicz functions; see Theorem 4.1. This theorem is crucial for verifying that the commutator $[M_\phi, T]$ (resp. Marcinkiewicz multiplier T_{m}) satisfies the weak Cesàro property in Sect. 5 (resp. Sect. 6).

In Sect. 5, we provide the proof of Theorem 1.2. Actually, with the help of Theorem 3.1, the main part of this section is occupied by showing that the commutator $[M_\phi, T]$ satisfies WCP(2); see Theorem 5.1. The optimal estimate in Theorem 1.2 is established in Sect. 5.2.

Finally, we prove Theorems 1.4 and 1.5 separately in Sect. 6.1 and 6.2. We explain in Remark 6.10 why elements $T_{\mathsf{m}} f$ and $S(f)$ have different distribution estimates.

Remark 1.8 We conjecture that the distributional estimates in Theorems 1.4 and 1.5 are optimal. However, at this moment, we are unable to prove this conjecture.

Throughout, we use c_A to denote some constant depending only on the symbol A, which may change from line to line. We use c_{abs} to denote an absolute constant which may also vary from line to line. For $1 \le p \le \infty$, denote by p' the conjugate index of p, that is, $p' = \frac{p}{p-1}$. For a given measurable function f on $(\Omega, \mathcal{F}, \nu)$, denote by $\mathrm{supp}(f)$ the support set of f. For each $A \in \mathcal{F}$, set $A^c = \Omega \backslash A$. For each Lebesgue measurable set A in Euclidean space $\mathbb{R}^d$, denote by $m(A)$ the Lebesgue measure of A. We denote the space of Schwartz functions on $\mathbb{R}^d$ by $\mathcal{S}(\mathbb{R}^d)$. For each cube $Q \subset \mathbb{R}^d$, c_Q denotes the center of Q, and rQ (here $r \in (0, \infty)$) is the dilation of Q with respect to its center.

2 Preliminaries

In this section, we introduce the definitions of symmetric quasi-Banach function spaces, (fractional) Cesàro operators. Also, we recall some properties of BMO space and Calderón–Zygmund decomposition.

2.1 Symmetric quasi-Banach function spaces

Let $(\Omega, \mathcal{F}, \nu)$ be a complete measure space. Denote by $L_0(\Omega, \mathcal{F}, \nu)$ (briefly $L_0(\Omega)$) the space of all measurable real-valued functions on the measure space $(\Omega, \mathcal{F}, \nu)$. For $f \in L_0(\Omega)$, the decreasing rearrangement of the function f (the right inverse of the distribution function of $|f|$) is defined by

$$\mu_{(\Omega,\nu)}(t, f) = \inf \{s > 0 : \nu(\{\xi \in \Omega : |f(\xi)| > s\}) \le t\}, \quad t > 0.$$

In the sequel, denote $\mu(t, f)$ for $\mu_{(\Omega,\nu)}(t, f)$ except for particular cases. For $f, g \in L_0(\Omega)$, we say that f is submajorized by g in the sense of Hardy–Littlewood–Polya (written by $f \prec\prec g$) whenever

$$\int_0^t \mu(s, f)ds \le \int_0^t \mu(s, g)ds, \quad \forall\, t > 0.$$

A Banach (or quasi-Banach) function space $(E(\Omega), \|\cdot\|_{E(\Omega)})$ on Ω is called symmetric if for every $g \in E$ and for every measurable function $f \in L_0(\Omega)$ with $\mu(f) \le \mu(g)$, we have $f \in E$ and $\|f\|_{E(\Omega)} \le \|g\|_{E(\Omega)}$. For the general theory of symmetric (quasi-) Banach functions spaces, we refer the reader to [6, 31, 32].

In the present paper, we will deal with the following examples of symmetric quasi-Banach function spaces (see e.g. [6, 31]).

- **Weak Lebesgue spaces.** For $0 < p < \infty$, the weak Lebesgue space $L_{p,\infty}(\Omega)$ is defined by

$$L_{p,\infty}(\Omega) = \left\{ f \in L_0(\Omega) : \sup_{\lambda>0} \lambda\nu\left(\{|f| > \lambda\}\right)^{1/p} < \infty \right\}$$

  equipped with the quasi-norm

$$\|f\|_{L_{p,\infty}(\Omega)} = \sup_{\lambda>0} \lambda\nu\left(\{|f| > \lambda\}\right)^{1/p} = \sup_{t>0} t^{1/p}\mu(t, f).$$

- **Lorentz spaces.** Let $\psi : [0, \infty) \to [0, \infty)$ be an increasing concave continuous function such that $\lim_{t\to\infty} \psi(t) = \infty$. The Lorentz space $\Lambda_\psi(\Omega)$ is defined by setting

$$\Lambda_\psi(\Omega) = \left\{ f \in L_0(\Omega) : \int_0^\infty \mu(s, f)d\psi(s) < \infty \right\}$$

  equipped with the norm

$$\|f\|_{\Lambda_\psi(\Omega)} = \int_0^\infty \mu(s, f)d\psi(s).$$

  We always denote by $\Lambda_{\log^\beta}(\Omega)$, $\beta > 0$, the Lorentz space $\Lambda_\psi(\Omega)$ with $\psi(t) = \log^\beta(e^{\beta+1} + t)$, $t \ge 0$.
- **Marcinkiewicz space.** Let ψ be as above. The Marcinkiewicz space M_ψ is defined by setting

$$\mathsf{M}_\psi(\Omega) = \left\{ f \in L_0(\Omega) : \int_0^\infty \mu(s, f)d\psi(s) < \infty \right\}$$

equipped with the norm

$$\|f\|_{\mathsf{M}_\psi(\Omega)} = \sup_{t>0} \frac{1}{\psi(t)} \int_0^t \mu(s, f)ds.$$

We always denote by $\mathsf{M}_{\log^\beta}(\Omega)$, $\beta > 0$, the Marcinkiewicz space $\mathsf{M}_\psi(\Omega)$ with $\psi(t) = \log^\beta(e^{\beta+1} + t)$, $t \geq 0$.

- **Orlicz spaces.** Recall that an Orlicz function Φ is a convex function such that $\Phi(0) = 0$ and $\Phi(\infty) = \infty$. The Orlicz function space $L_\Phi(\Omega)$ is the set of all measurable functions f on Ω such that

$$\|f\|_{L_\Phi(\Omega)} = \inf\left\{\lambda > 0 : \int_\Omega \Phi\left(\frac{|f(\xi)|}{\lambda}\right) d\nu(\xi) \leq 1\right\} < \infty.$$

Given a symmetric Banach function space $E(\Omega)$, the Köthe dual of $E(\Omega)$ is defined as

$$E(\Omega)^\times = \left\{f \in L_0(\Omega) : |\int_\Omega fgd\nu| < \infty \text{ for all } g \in E(\Omega)\right\}$$

equipped with the norm

$$\|f\|_{E(\Omega)^\times} = \sup\left\{|\int_\Omega fgd\nu| : g \in E(\Omega),\ \|g\|_{E(\Omega)} \leq 1\right\}.$$

If Banach spaces F_1 and F_2 are linear subspaces of a Hausdorff topological vector space, then the intersection $F_1 \cap F_2$ becomes a Banach space when equipped with the norm

$$\|f\|_{F_1\cap F_2} = \max\left\{\|f\|_{F_1}, \|f\|_{F_2}\right\}.$$

The sum $F_1 + F_2$ becomes a Banach space when equipped with the norm

$$\|f\|_{F_1+F_2} = \inf\left\{\|f_1\|_{F_1} + \|f_2\|_{F_2} : f = f_1 + f_2, f_1 \in F_1, f_2 \in F_2\right\} < \infty.$$

The above facts can be found in [7, Lemma 2.3.1].

The following well-known lemmas will be used several times in the sequel. One can find them in [6] or [31, Chapter II] or [25] or [32, Proposition 2.a.2].

Lemma 2.1 *Let $\psi : [0, \infty) \to [0, \infty)$ be an increasing concave continuous function such that $\psi(0) = 0$ and $\lim_{t\to\infty} \psi(t) = \infty$. We have*

$$\Lambda_\psi(\Omega)^\times = \mathsf{M}_\psi(\Omega), \qquad \mathsf{M}_\psi(\Omega)^\times = \Lambda_\psi(\Omega)$$

with equality of the norms.

Lemma 2.2 *For each $f \in (L_1 + L_\infty)(\Omega)$, we have*

$$\|f\|_{(L_1+L_\infty)(\Omega)} = \int_0^1 \mu(s, f)ds.$$

2.2 BMO spaces and John–Nirenberg inequalities

Denote by $L_1^{\rm loc}(\mathbb{R}^d)$ the space of all locally integrable functions on $\mathbb{R}^d$. Recall that the $\mathrm{BMO}(\mathbb{R}^d)$ space, the class of functions of bounded mean oscillations, is defined by setting

$$\|\phi\|_{\rm BMO} = \sup_Q \frac{1}{|Q|}\int_Q |\phi(x) - \phi_Q|dx, \quad \phi \in L_1^{\rm loc}(\mathbb{R}^d),$$

where the supremum is taken over all cubes Q in $\mathbb{R}^d$, and

$$\phi_Q = \frac{1}{|Q|}\int_Q \phi(x)dx.$$

We need the following result which is a consequence of the John-Nirenberg inequality.

Lemma 2.3 *Let $\Phi^*(t) = e^t - t - 1$ for each $t \geq 0$. For each cube $Q \subset \mathbb{R}^d$, we have*

$$\|\phi - \phi_Q\|_{L_{\Phi^*}(Q, \frac{m}{m(Q)})} \leq 2^{d+2}e^2\|\phi\|_{\mathrm{BMO}(\mathbb{R}^d)}, \quad \phi \in \mathrm{BMO}(\mathbb{R}^d).$$

Proof Without loss of generality, assume $\|\phi\|_{\mathrm{BMO}(\mathbb{R}^d)} = (2^{d+2}e)^{-1}$. The John-Nirenberg inequality (see e.g. Theorem 7.1.6 in [20]) asserts that

$$m(\{x \in Q : |\phi(x) - \phi_Q| > t\}) \leq e^{1-4t}m(Q), \quad t > 0.$$

Thus, we have

$$\begin{aligned}\int_Q e^{|\phi(x)-\phi_Q|}dx &= -\int_0^\infty e^t d\left(m\left(\{x \in Q : |\phi(x) - \phi_Q| > t\}\right)\right)\\ &\leq -m(Q)\cdot\int_0^\infty e^t d\left(\min\left\{e^{1-4t}, 1\right\}\right)\\ &= -m(Q)\cdot\int_{\frac{1}{4}}^\infty e^t d(e^{1-4t}) = \frac{4}{3}e^{\frac{1}{4}}m(Q).\end{aligned}$$

Thus,

$$\frac{1}{m(Q)}\int_Q \Phi^*(|\phi(x) - \phi_Q|)dx \leq \frac{1}{m(Q)}\int_Q e^{|\phi(x)-\phi_Q|}dx - 1 \leq \frac{4}{3}e^{\frac{1}{4}} - 1 \leq 1.$$

In other words, $\|\phi - \phi_Q\|_{L_{\Phi^*}(Q, \frac{m}{m(Q)})} \leq 1$. □

2.3 Dyadic Calderón–Zygmund decomposition

For each $k \in \mathbb{Z}$, denote by $\mathcal{D}_k$ the set of all dyadic cubes in $\mathbb{R}^d$ whose side length is 2^{-k}. For each $k \in \mathbb{Z}$, set $\mathcal{F}_k = \sigma(\mathcal{D}_k)$ the σ-algebra generated by $\mathcal{D}_k$, and set

$$\mathcal{D} = \bigcup_{k\in\mathbb{Z}} \mathcal{D}_k.$$

It is clear that the conditional expectation of $f \in L_1^{\mathrm{loc}}(\mathbb{R}^d)$ with respect to $\mathcal{F}_k$, $k \in \mathbb{Z}$, is given by

$$E_k(f) = \sum_{Q\in\mathcal{D}_k} f_Q \chi_Q. \tag{2.1}$$

The dyadic Hardy–Littlewood maximal function is given by

$$M_{\mathcal{D}} f = \sup_{k\in\mathbb{Z}} |E_k(f)|.$$

Note that this maximal function is also the Doob maximal function in the martingale setting [26, 33, 45]. In the rest of this subsection, we recall the dyadic Calderón–Zygmund decomposition. For the reader's convenience, we also include some basic estimates.

Construction 2.4 *Let $\lambda > 0$ and let $0 \le f \in (L_1 \cap L_\infty)(\mathbb{R}^d)$. Set*

$$\Omega_\lambda = \left\{ x \in \mathbb{R}^d : \ M_{\mathcal{D}} f(x) > \lambda \right\},$$

and

$$\Omega_{n,\lambda} = \left\{ x \in \mathbb{R}^d : \ (E_n f)(x) > \lambda, \quad (E_m f)(x) \le \lambda \text{ for every } m < n \right\}, \quad n \in \mathbb{Z}.$$

Clearly,

$$\Omega_\lambda = \bigcup_{n\in\mathbb{Z}} \Omega_{n,\lambda}.$$

Since each $\Omega_{n,\lambda} \in \mathcal{F}_n$, it follows that there exists $\mathcal{Q}_n \subset \mathcal{D}_n$ such that

$$\Omega_{n,\lambda} = \bigcup_{Q\in\mathcal{Q}_n} Q.$$

Set $\mathcal{Q} = \cup_{n\in\mathbb{Z}} \mathcal{Q}_n$ and denote

$$g = f\chi_{\Omega_\lambda^c} + \sum_{Q\in\mathcal{Q}} f_Q \chi_Q, \quad b = \sum_{Q\in\mathcal{Q}} (f - f_Q)\chi_Q.$$

This means that Ω_λ is the disjoint union of cubes in $\mathcal{Q}$.

Lemma 2.5 *Let $\lambda > 0$, Ω_λ, f and g be as in Construction 2.4. We have*

(i) $m(\Omega_\lambda) \le \lambda^{-1}\|f\|_{L_1(\mathbb{R}^d)}$.
(ii) $\|g\|^2_{L_2(\mathbb{R}^d)} \le 2^d\lambda\|f\|_{L_1(\mathbb{R}^d)}$.

Proof (i) We have

$$\lambda m(\Omega_\lambda) = \sum_{Q\in\mathcal{Q}} \lambda m(Q) \le \sum_{Q\in\mathcal{Q}} f_Q m(Q) = \int_{\Omega_\lambda} f(x)dx \le \|f\|_{L_1(\mathbb{R}^d)}.$$

(ii) If $Q \in \mathcal{Q}$, then $Q \subset \Omega_{n,\lambda}$ for some $n \in \mathbb{Z}$. Hence, $E_n f > \lambda$ and $E_{n-1} f \le \lambda$ on Q. That is, $f_Q > \lambda$ and $f_{\widehat{Q}} \le \lambda$ (here, $\widehat{Q}$ is the dyadic parent cube for Q). Since $f \ge 0$, it follows that $f_Q \le 2^d f_{\widehat{Q}} \le 2^d\lambda$. Also, $f \le M_{\mathcal{D}} f \le \lambda$ on Ω^c_λ. Hence,

$$\begin{aligned}\|g\|^2_{L_2(\mathbb{R}^d)} &= \int_{\Omega^c_\lambda} f(x)^2dx + \sum_{Q\in\mathcal{Q}} f_Q^2 m(Q) \le \lambda\int_{\Omega^c_\lambda} f(x)dx + 2^d\lambda\sum_{Q\in\mathcal{Q}} f_Q m(Q)\\ &= \lambda\int_{\Omega^c_\lambda} f(x)dx + 2^d\lambda\int_{\Omega_\lambda} f(x)dx \le 2^d\lambda\|f\|_{L_1(\mathbb{R}^d)}.\end{aligned}$$

□

2.4 Cesàro operator and its fractional version

The Cesàro operator

$$\mathscr{C} : (L_1 + L_\infty)(0,\infty) \to (L_{1,\infty} + L_\infty)(0,\infty)$$

is defined by

$$\mathscr{C} f(t) = \frac{1}{t}\int_0^t f(s)ds, \quad f \in (L_1 + L_\infty)(0,\infty).$$

The dual of $\mathscr{C}$ (in the Hilbert space sense), $\mathscr{C}^* : \Lambda_{\log}(0,\infty) \to L_0(0,\infty)$ is given by

$$\mathscr{C}^* f(t) = \int_t^\infty \frac{f(s)}{s}ds, \quad f \in \Lambda_{\log}(0,\infty).$$

An important and well-known fact of the Cesàro operator is stated as follows (see e.g. [22, Theorem 327]):

$$\|\mathscr{C}\|_{L_p(0,\infty)\to L_p(0,\infty)} = \frac{p}{p-1}, \qquad 1 < p \le \infty, \tag{2.2}$$

and

$$\|\mathscr{C}^*\|_{L_p(0,\infty)\to L_p(0,\infty)} = p, \qquad 1 \le p < \infty. \tag{2.3}$$

In this paper, we also need the notion of fractional power of Cesàro operators $\mathscr{C}$ and $\mathscr{C}^*$. For each $0 < \alpha < \infty$, let $\mathscr{C}^\alpha : (L_1 + L_\infty)(0,\infty) \to L_0(0,\infty)$ be given by

$$\mathscr{C}^\alpha f(t) = \frac{1}{\Gamma(\alpha)} \frac{1}{t} \int_0^t f(s) \log^{\alpha-1}\left(\frac{t}{s}\right) ds. \tag{2.4}$$

Its dual operator, $(\mathscr{C}^*)^\alpha : \Lambda_{\log^\alpha}(0,\infty) \to L_0(0,\infty)$, is given by

$$(\mathscr{C}^*)^\alpha f(t) = \frac{1}{\Gamma(\alpha)} \int_t^\infty f(s) \log^{\alpha-1}\left(\frac{s}{t}\right) \frac{ds}{s} \tag{2.5}$$

where $\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t} dt$ is the Gamma function. When $\alpha = 1$, we have $\mathscr{C}^\alpha = \mathscr{C}$ and $(\mathscr{C}^*)^\alpha = \mathscr{C}^*$. The operator $(\mathscr{C}^*)^\alpha$ was recently introduced and studied in [27]. More precisely, it was proved in [27, Lemma 3.1] and [27, Lemma 3.2] that, for each $1 \le p < \infty$,

$$\|(\mathscr{C}^*)^\alpha\|_{L_p(0,\infty)\to L_p(0,\infty)} \le p^\alpha, \quad \alpha > 0 \tag{2.6}$$

and therefore $\{(\mathscr{C}^*)^\alpha\}_\alpha$ is a bounded linear operator semigroup on $L_p(0,\infty)$ $(1 \le p < \infty)$ in the sense that

$$(\mathscr{C}^*)^\alpha \circ (\mathscr{C}^*)^\beta = (\mathscr{C}^*)^{\alpha+\beta}, \quad \alpha, \beta > 0. \tag{2.7}$$

Applying (2.6) and (2.7), we can show that, for each $1 < p \le \infty$,

$$\|\mathscr{C}^\alpha\|_{L_p(0,\infty)\to L_p(0,\infty)} \le \left(\frac{p}{p-1}\right)^\alpha, \quad \alpha > 0. \tag{2.8}$$

and $\{\mathscr{C}^\alpha\}_\alpha$ is a bounded linear operator semigroup on $L_p(0,\infty)$ $(1 < p \le \infty)$ in the sense that

$$\mathscr{C}^\alpha \circ \mathscr{C}^\beta = \mathscr{C}^{\alpha+\beta}, \quad \alpha, \beta > 0. \tag{2.9}$$

It is interesting to compare (2.6), (2.8) with (2.2), (2.5).

For $f \in L_0(0,\infty)$, the dilation operator σ_s associated with $s > 0$ is defined by

$$(\sigma_s f)(t) = f\left(\frac{t}{s}\right).$$

It is not hard to check that the dilation operator commutes with both $\mathscr{C}^\alpha$ and $(\mathscr{C}^*)^\beta$.

3 An abstract extrapolation theorem

In this section, we establish Theorem 3.1, an abstract extrapolation theorem, which will help us to prove the pointwise distribution estimates, Theorems 1.2, 1.4 and 1.5.

We always assume that $(\Omega, \mathcal{F}, \nu)$ is a complete σ-finite measure space. In the proof of Theorem 3.1, we shall need the concave increasing and continuous function $\psi_{\alpha,\beta}$, $\alpha, \beta > 0$, given by the formula

$$\psi_{\alpha,\beta}(t) = \begin{cases} \frac{1}{\Gamma(\alpha+1)} \int_0^t \log^{\alpha}(\frac{1}{s})ds + t, & t \in (0,1], \\ 2 + \frac{1}{\Gamma(\beta)} \int_1^t (\int_0^{s^{-1}} \log^{\beta-1}(\frac{1}{u})du)ds, & t \geq 1. \end{cases} \tag{3.1}$$

Theorem 3.1 *Let $\alpha, \beta > 0$. Let $T : L_2(\Omega, \mathcal{F}, \nu) \to L_2(\Omega, \mathcal{F}, \nu)$ be a linear contraction. Suppose that T admits a bounded linear extension on $L_p(\Omega, \mathcal{F}, \nu)$ for all $1 < p < \infty$.*

(i) *If*

$$\|T\|_{L_p(\Omega,\mathcal{F},\nu)\to L_p(\Omega,\mathcal{F},\nu)} \leq \begin{cases} (p')^{\alpha}, & 1 < p \leq 2 \\ p^{\beta}, & 2 \leq p < \infty \end{cases}, \tag{3.2}$$

then there exists a positive constant $c_{\alpha,\beta}$ such that

$$\mu(Tf) \prec\prec c_{\alpha,\beta}(\mathscr{C}^{\alpha} + (\mathscr{C}^*)^{\beta})\mu(f)$$

for every $f \in \Lambda_{\psi_{\alpha,\beta}}(\Omega, \mathcal{F}, \nu)$.

(ii) *Let T satisfy conditions of* (i) *and additional $WCP(\alpha)$, i.e, for every $f \in L_0(\Omega, \mathcal{F}, \nu)$ supported on a set of finite measure and for every $\lambda \geq (\mathscr{C}^{\alpha}\mu(f))(\nu(\mathrm{supp}(f)))$,*

$$\lambda\nu\Big(\{x \in \Omega : |Tf|(x) > \lambda\}\Big) \leq \nu(\mathrm{supp}(f)) \cdot (\mathscr{C}^{\alpha}\mu(f))(\nu(\mathrm{supp}(f))). \tag{3.3}$$

There exists a positive constant $c'_{\alpha,\beta}$ such that

$$\mu(Tf) \leq c'_{\alpha,\beta}(\mathscr{C}^{\alpha} + (\mathscr{C}^*)^{\beta})\mu(f)$$

for each $f \in L_0(\Omega)$ with $\mu(f) \in \mathrm{dom}(\mathscr{C}^{\alpha}) \cap \mathrm{dom}((\mathscr{C}^*)^{\beta})$.

Remark 3.2 Theorem 3.1 implies Yano's extrapolation theorem [46]. It is also easy to see that if $\alpha = \beta = 1$, then condition (3.3) implies that T is of weak type $(1, 1)$. Thus, Theorem 3.1 extends [43, Theorem 14], [27, Theorem 4.1] and [28, Theorem 3.9].

3.1 Proof of Theorem 3.1 (i)

In this subsection, we prove the first assertion of Theorem 3.1. We need several lemmas. Recall that the Lorentz spaces $\Lambda_{\log^\beta}(\Omega)$ and Marcinkiewicz spaces $\mathsf{M}_{\log^\beta}(\Omega)$ are defined by the function $\psi(t) = \log^\beta(e^{\beta+1} + t)$, $t \geq 0$, in Sect. 2.

Lemma 3.3 *Let $\beta > 0$. If $f \in \bigcap_{1<p\leq 2} L_p(0,\infty)$ is such that*

$$\sup_{1<p\leq 2} (p')^{-\beta} \|f\|_{L_p(0,\infty)} < \infty,$$

then

$$\|f\|_{\mathsf{M}_{\log^\beta}(0,\infty)} \leq 2^{\beta+2} \sup_{1<p\leq 2} (p')^{-\beta} \|f\|_{L_p(0,\infty)}.$$

Proof Without loss of generality, assume

$$\sup_{1<p\leq 2} (p')^{-\beta} \|f\|_{L_p(0,\infty)} = 1.$$

By Hölder's inequality, for each $t \in (0,\infty)$, we have

$$\int_0^t \mu(s,f)ds \leq \|f\|_{L_p(0,\infty)} \|\chi_{(0,t)}\|_{L_{p'}(0,\infty)} \leq (p')^\beta t^{\frac{1}{p'}}, \quad \text{for all} \quad 1 < p \leq 2,$$

where p' is the conjugate index number of p. If $t \geq e^2$, then set $p' = \log(t) \in [2,\infty)$. Therefore, we have

$$\int_0^t \mu(s,f)ds \leq (\log(t))^\beta \cdot t^{\frac{1}{\log(t)}} = e\log^\beta(t) \leq 2^{\beta+2}\log^\beta(e^{\beta+1}+t).$$

If $t \leq e^2$, then $t \leq 8$ and

$$\int_0^t \mu(s,f)ds \leq \int_0^8 \mu(s,f)ds \leq 2\sqrt{2}\|f\|_{L_2(0,\infty)} \leq 2^{\beta+2} \leq 2^{\beta+2}\log^\beta(e^{\beta+1}+t).$$

These two estimates together with the definition of $\mathsf{M}_{\log^\beta}(\Omega)$ give us the desired assertion. □

Lemma 3.4 *Let $\beta > 0$, and let linear contraction $T : L_2(\Omega,\mathcal{F},\nu) \to L_2(\Omega,\mathcal{F},\nu)$ be such that*

$$\|T(f)\|_{L_p(\Omega,\mathcal{F},\nu)} \leq p^\beta \|f\|_{L_p(\Omega,\mathcal{F},\nu)}, \quad f \in (L_p \cap L_2)(\Omega,\mathcal{F},\nu), \quad 2 \leq p < \infty.$$

We have

$$\|T(f)\|_{(L_2+L_\infty)(\Omega,\mathcal{F},\nu)} \leq 2^{\beta+2}\|f\|_{\Lambda_{\log^\beta}(\Omega,\mathcal{F},\nu)}, \quad f \in L_2(\Omega,\mathcal{F},\nu).$$

Proof By duality, we write

$$\|T(f)\|_{(L_2+L_\infty)(\Omega,\mathcal{F},\nu)} = \sup_{\substack{g\in L_\infty(\Omega,\mathcal{F},\nu)\\ \nu(\mathrm{supp}(g))<\infty\\ \|g\|_{(L_1\cap L_2)(\Omega,\mathcal{F},\nu)}\le 1}} |\langle T(f), g\rangle_{L_2(\Omega,\mathcal{F},\nu)}|.$$

We proceed to write (as usual, p' denotes the conjugate index of p)

$$\begin{aligned}|\langle T(f), g\rangle_{L_2(\Omega,\mathcal{F},\nu)}| &= |\langle f, T^*(g)\rangle_{L_2(\Omega,\mathcal{F},\nu)}|\\ &\le \|f\|_{\Lambda_{\log^\beta}(\Omega,\mathcal{F},\nu)}\|T^*(g)\|_{\mathsf{M}_{\log^\beta}(\Omega,\mathcal{F},\nu)}\\ &\le 2^{\beta+2}\|f\|_{\Lambda_{\log^\beta}(\Omega,\mathcal{F},\nu)}\cdot \sup_{1<p\le 2}(p')^{-\beta}\|T^*(g)\|_{L_p(\Omega,\mathcal{F},\nu)},\end{aligned}$$

where we use Lemma 2.1 in the first inequality and Lemma 3.3 in the second inequality. Moreover, the assumptions on T imply that

$$\begin{aligned}&\sup_{1<p\le 2}(p')^{-\beta}\|T^*(g)\|_{L_p(\Omega,\mathcal{F},\nu)}\\ &\quad\le \sup_{1<p\le 2}(p')^{-\beta}\|T^*\|_{L_p(\Omega,\mathcal{F},\nu)\to L_p(\Omega,\mathcal{F},\nu)}\cdot \sup_{1<p\le 2}\|g\|_{L_p(\Omega,\mathcal{F},\nu)}\\ &\quad= \sup_{1<p\le 2}(p')^{-\beta}\|T\|_{L_{p'}(\Omega,\mathcal{F},\nu)\to L_{p'}(\Omega,\mathcal{F},\nu)}\cdot \sup_{1<p\le 2}\|g\|_{L_p(\Omega,\mathcal{F},\nu)}\\ &\quad\le \sup_{1<p\le 2}\|g\|_{L_p(\Omega,\mathcal{F},\nu)} = \|g\|_{(L_1\cap L_2)(\Omega,\mathcal{F},\nu)}.\end{aligned}$$

Combining these estimates, we complete the proof. □

Lemma 3.5 *Let T, α and β be as in Theorem 3.1 (i). There exists $c_{\alpha,\beta}>0$ such that*

$$\|Tf\|_{(L_1+L_\infty)(\Omega,\mathcal{F},\nu)} \le c_{\alpha,\beta}\|f\|_{\Lambda_{\psi_{\alpha,\beta}}(\Omega,\mathcal{F},\nu)}$$

for each function f taking finitely many values and supported on a set of finite measure.

Proof Consider a convex functional defined on all functions taking finitely many values and supported on a set of finite measure. If this functional is bounded (in the space $\Lambda_{\psi_{\alpha,\beta}}(\Omega,\mathcal{F},\nu)$) on all indicator functions, then it is also bounded (in the space $\Lambda_{\psi_{\alpha,\beta}}(\Omega,\mathcal{F},\nu)$) on functions taking finitely many values and supported on a set of finite measure (see e.g. Lemma 5.2 in [31, p. 111]). We apply the argument above to the convex functional

$$f\to \|Tf\|_{(L_1+L_\infty)(\Omega,\mathcal{F},\nu)}.$$

Thus, it suffices to show that there exists a positive constant $c_{\alpha,\beta}$ such that

$$\|T(\chi_A)\|_{(L_1+L_\infty)(\Omega,\mathcal{F},\nu)} \le c_{\alpha,\beta}\psi_{\alpha,\beta}(\nu(A)),\quad A\subset\Omega.$$

Case 1. Suppose first that $\nu(A) = t \in [0, e^{-2}]$. For every $p \in (1, 2]$, we have

$$\|T(\chi_A)\|_{(L_1+L_\infty)(\Omega,\mathcal{F},\nu)} \le \|T(\chi_A)\|_{L_p(\Omega,\mathcal{F},\nu)} \le (p')^\alpha \|\chi_A\|_{L_p(\Omega,\mathcal{F},\nu)} = (p')^\alpha t^{1-\frac{1}{p'}}.$$

Take $p' = \log(\frac{1}{t}) \ge 2$. We have $t^{-\frac{1}{p'}} = e$, and therefore

$$\|T(\chi_A)\|_{(L_1+L_\infty)(\Omega,\mathcal{F},\nu)} \le et \log^\alpha\left(\frac{1}{t}\right).$$

Case 2: Suppose now $\nu(A) = t \in [e^{-2}, 1]$. We have

$$\|T(\chi_A)\|_{(L_1+L_\infty)(\Omega,\mathcal{F},\nu)} \le \|T(\chi_A)\|_{L_2(\Omega,\mathcal{F},\nu)} \le \|\chi_A\|_{L_2(\Omega,\mathcal{F},\nu)} \le 1.$$

Case 3. Suppose now that $\nu(A) = t \in [1, \infty)$. Using Lemma 3.4, we write

$$\begin{aligned}\|T(\chi_A)\|_{(L_1+L_\infty)(\Omega,\mathcal{F},\nu)} &\le \|T(\chi_A)\|_{(L_2+L_\infty)(\Omega,\mathcal{F},\nu)} \\ &\le 2^{\beta+2}\|\chi_A\|_{\Lambda_{\log^\beta}(\Omega,\mathcal{F},\nu)} = 2^{\beta+2}\log^\beta(e^{\beta+1}+t).\end{aligned}$$

The proof is complete. □

Lemma 3.6 *Let $\alpha, \beta > 0$, and let $\psi_{\alpha,\beta}$ be as in (3.1). For each $f \in \Lambda_{\psi_{\alpha,\beta}}(\Omega, \mathcal{F}, \nu)$, we have*

$$\|f\|_{\Lambda_{\psi_{\alpha,\beta}}(\Omega,\mathcal{F},\nu)} = \|(\mathscr{C}^\alpha + (\mathscr{C}^*)^\beta)\mu(f)\|_{(L_1+L_\infty)(0,\infty)}.$$

Proof By (2.2), and since $\mathscr{C}$ and $\mathscr{C}^*$ send decreasing functions to decreasing functions, we have

$$\|(\mathscr{C}^\alpha + (\mathscr{C}^*)^\beta)\mu(f)\|_{(L_1+L_\infty)(0,\infty)} = \int_0^1 \mathscr{C}^\alpha\mu(f)(t)dt + \int_0^1 (\mathscr{C}^*)^\beta\mu(f)(t)dt.$$

Note that the identity $\int_0^s \log^{\beta-1}(\frac{s}{t})\frac{dt}{s} = \Gamma(\beta)$. For the above two terms on the right-hand side, we have

$$\begin{aligned}\int_0^1 \mathscr{C}^\alpha\mu(f)(t)dt &= \frac{1}{\Gamma(\alpha)}\int_0^1 \frac{1}{t}\left(\int_0^t \mu(s,f)\log^{\alpha-1}\left(\frac{t}{s}\right)ds\right)dt \\ &= \frac{1}{\Gamma(\alpha)}\int_0^1 \mu(s,f)\left(\int_s^1 \log^{\alpha-1}\left(\frac{t}{s}\right)\frac{dt}{t}\right)ds \\ &= \frac{1}{\Gamma(\alpha+1)}\int_0^1 \mu(s,f)\log^\alpha\left(\frac{1}{s}\right)ds\end{aligned}$$

and

$$\int_0^1 (\mathscr{C}^*)^\beta\mu(f)(t)dt = \frac{1}{\Gamma(\beta)}\int_0^1 \left(\int_t^\infty \mu(s,f)\log^{\beta-1}\left(\frac{s}{t}\right)\frac{ds}{s}\right)dt$$

$$= \frac{1}{\Gamma(\beta)} \int_0^\infty \mu(s, f) \left(\int_0^{\min\{s,1\}} \log^{\beta-1}(\frac{s}{t}) dt \right) \frac{ds}{s}$$
$$= \int_0^1 \mu(s, f) ds + \frac{1}{\Gamma(\beta)} \int_1^\infty \mu(s, f) \left(\int_0^{s^{-1}} \log^{\beta-1} \left(\frac{1}{u} \right) du \right) ds.$$

The assertion follows now from the definition of $\Lambda_{\psi_{\alpha,\beta}}$. □

We are now fully equipped to prove Theorem 3.1 (i).

Proof of Theorem 3.1 (i) Let $f \in \Lambda_{\psi_{\alpha,\beta}}(\Omega, \mathcal{F}, \nu)$. For each $t > 0$, we have

$$\frac{1}{t} \int_0^t \mu(s, Tf) ds = \|\sigma_{\frac{1}{t}} \mu(Tf)\|_{(L_1+L_\infty)(0,\infty)}$$
$$= \|\mu_{(\Omega,t^{-1}\nu)}(Tf)\|_{(L_1+L_\infty)(0,\infty)} = \|Tf\|_{(L_1+L_\infty)(\Omega,\mathcal{F},t^{-1}\nu)},$$

where the first equality is due to Lemma 2.2. By the assumption on T, we have

$$\|Tf\|_{L_p(\Omega,\mathcal{F},t^{-1}\nu)} \le c_p \|f\|_{L_p(\Omega,\mathcal{F},t^{-1}\nu)}, \quad f \in L_p(\Omega, \mathcal{F}, \nu), \quad 1 < p < \infty$$

where $c_p = (p')^\alpha$ when $1 < p \le 2$ and $c_p = p^\beta$ when $2 \le p < \infty$. Hence, the assumptions of T in Lemma 3.5 are met for the measure $t^{-1}\nu$. Using Lemmas 3.5 and 3.6, we obtain

$$\begin{aligned}
\|Tf\|_{(L_1+L_\infty)(\Omega,\mathcal{F},t^{-1}\nu)} &\le c_{\alpha,\beta} \|f\|_{\Lambda_{\psi_{\alpha,\beta}}(\Omega,\mathcal{F},t^{-1}\nu)} \\
&= c_{\alpha,\beta} \|\mu_{(\Omega,\mathcal{F},t^{-1}\nu)}(f)\|_{\Lambda_{\psi_{\alpha,\beta}}(0,\infty)} \\
&= c_{\alpha,\beta} \|\sigma_{t^{-1}} \mu(f)\|_{\Lambda_{\psi_{\alpha,\beta}}(0,\infty)} \\
&= c_{\alpha,\beta} \|(\mathscr{C}^\alpha + (\mathscr{C}^*)^\beta) \sigma_{t^{-1}} \mu(f)\|_{(L_1+L_\infty)(0,\infty)} \\
&= c_{\alpha,\beta} \|\sigma_{t^{-1}} (\mathscr{C}^\alpha + (\mathscr{C}^*)^\beta) \mu(f)\|_{(L_1+L_\infty)(0,\infty)} \\
&= \frac{c_{\alpha,\beta}}{t} \int_0^t (\mathscr{C}^\alpha + (\mathscr{C}^*)^\beta) \mu(f)(s) ds
\end{aligned}$$

for every function f taking finitely many values and supported on a set of finite measure. Here, we also used the fact that the dilation operator commutes with the operators $\mathscr{C}^\alpha$, $(\mathscr{C}^*)^\beta$, and Lemma 2.2 in the last two equalities. Since $t > 0$ is arbitrary, it follows from the above argument that

$$Tf \prec\prec c_{\alpha,\beta} (\mathscr{C}^\alpha + (\mathscr{C}^*)^\beta) \mu(f)$$

for every function f taking finitely many values and supported on a set of finite measure. In particular, T extends to a bounded mapping from $\Lambda_{\psi_{\alpha,\beta}}(\Omega, \mathcal{F}, \nu)$ to

$(L_1 + L_\infty)(\Omega, \mathcal{F}, \nu)$ and

$$Tf \prec\prec c_{\alpha,\beta}(\mathscr{C}^\alpha + (\mathscr{C}^*)^\beta)\mu(f), \quad f \in \Lambda_{\psi_{\alpha,\beta}}(\Omega, \mathcal{F}, \nu).$$

We complete the proof. □

3.2 Proof of Theorem 3.1 (ii)

In this subsection, we prove the second assertion of Theorem 3.1. Let us begin with the following basic estimate whose proof is straightforward and is, therefore, omitted.

Lemma 3.7 *Let $\alpha, \beta > 0$. There exists a positive constant $c^{(1)}_{\alpha,\beta}$ such that*

$$\Big(\mathscr{C}^\alpha\mu(f) + (\mathscr{C}^*)^\beta\mu(f)\Big)(t) \le c^{(1)}_{\alpha,\beta}\Big(\mathscr{C}^\alpha\mu(f) + (\mathscr{C}^*)^\beta\mu(f)\Big)(2t), \quad t > 0,$$

for each $f \in L_0(\Omega)$ with $\mu(f) \in \mathrm{dom}(\mathscr{C}^\alpha) \cap \mathrm{dom}((\mathscr{C}^*)^\beta)$.

We need two more lemmas.

Lemma 3.8 *Let T satisfy property $WCP(\alpha)$. Given $t > 0$, let $f \in L_1(\Omega)$ be such that $\nu(\mathrm{supp}(f)) = t$. We have*

$$\mu(t, Tf) \le 2\mathscr{C}^\alpha\mu(f)(t).$$

Proof By the assumption, we have

$$\lambda\nu\Big(\{x \in \Omega : |Tf|(x) > \lambda\}\Big) \le t\mathscr{C}^\alpha\mu(f)(t)$$

for every $\lambda \ge \mathscr{C}^\alpha\mu(f)(t)$. Set

$$B_t = \{x \in \Omega : |Tf|(x) > \mathscr{C}^\alpha\mu(f)(t)\}.$$

When $0 < \lambda < \mathscr{C}^\alpha\mu(f)(t)$, we have

$$\begin{aligned}
&\lambda\nu\Big(\{x \in \Omega : |T(f)\chi_{B_t}|(x) > \lambda\}\Big)\\
&\quad = \lambda\nu\Big(\{x \in \Omega : |T(f)\chi_{B_t}|(x) > \mathscr{C}^\alpha\mu(f)(t)\}\Big)\\
&\quad \le \mathscr{C}^\alpha\mu(f)(t) \cdot \nu\Big(\{x \in \Omega : |Tf|(x) > \mathscr{C}^\alpha\mu(f)(t)\}\Big)\\
&\quad \le t\mathscr{C}^\alpha\mu(f)(t).
\end{aligned}$$

Therefore,

$$\lambda\nu\Big(\{x \in \Omega : |T(f)\chi_{B_t}|(x) > \lambda\}\Big) \le t\mathscr{C}^\alpha\mu(f)(t)$$

for every $\lambda > 0$. In other words,

$$\|T(f)\chi_{B_t}\|_{L_{1,\infty}(\Omega,\mathcal{F},\nu)} \le t\mathscr{C}^{\alpha}\mu(f)(t).$$

Consequently,

$$\begin{aligned}\mu(t, T(f)) &\le \mu(t, T(f)\chi_{B_t}) + \mu(0, T(f)\chi_{\Omega\setminus B_t})\\ &\le \frac{1}{t}\|T(f)\chi_{B_t}\|_{L_{1,\infty}(\Omega,\mathcal{F},\nu)} + \|T(f)\chi_{\Omega\setminus B_t}\|_{L_\infty(\Omega,\mathcal{F},\nu)}\\ &\le \frac{1}{t}\cdot t\mathscr{C}^{\alpha}\mu(f)(t) + \mathscr{C}^{\alpha}\mu(f)(t) \le 2\mathscr{C}^{\alpha}\mu(f)(t).\end{aligned}$$

This is the desired result. □

Lemma 3.9 *Suppose we are in the setting of Theorem 3.1 (ii). There exists a positive constant $c'_{\alpha,\beta}$ such that*

$$\mu(Tf)\chi_{(0,2\nu(\mathrm{supp}(f)))} \le c'_{\alpha,\beta}(\mathscr{C}^{\alpha} + (\mathscr{C}^{*})^{\beta})\mu(f)$$

for every $f \in L_0(\Omega)$ such that $\mu(f) \in \mathrm{dom}(\mathscr{C}^{\alpha}) \cap \mathrm{dom}((\mathscr{C}^{*})^{\beta})$.

Proof Fix $0 < t < \nu(\mathrm{supp}(f))$ and decompose $f = f_1 + f_2$ such that

$$\mu(u, f_1) = \mu(u, f)\chi_{(0,t)}(u), \quad \mu(u, f_2) = \mu(u + t, f), \quad u > 0.$$

Note that $\nu(\mathrm{supp}(f_1)) = t$. By Lemma 3.8, we have

$$\mu(t, T(f_1)) \le 2\mathscr{C}^{\alpha}\mu(f_1)(t) \le 2\mathscr{C}^{\alpha}\mu(f)(t). \tag{3.4}$$

Since $\mu(f) \in \mathrm{dom}(\mathscr{C}^{\alpha})\cap\mathrm{dom}((\mathscr{C}^{*})^{\beta})$, we have $\mu(f_2) \in \mathrm{dom}(\mathscr{C}^{\alpha})\cap\mathrm{dom}((\mathscr{C}^{*})^{\beta})$. Applying Theorem 3.1 (i) to T and f_2, we infer

$$t\mu(t, T(f_2)) \le \int_0^t \mu(s, T(f_2))ds \le c_{\alpha,\beta}\int_0^t (\mathscr{C}^{\alpha} + (\mathscr{C}^{*})^{\beta})\mu(f_2)(s)ds. \tag{3.5}$$

Clearly, $\mu(f_2) \le \min\{\mu(f), \mu(t, f)\}$. Hence,

$$\int_0^t (\mathscr{C}^{\alpha} + (\mathscr{C}^{*})^{\beta})\mu(f_2)(s)ds \le \int_0^t (\mathscr{C}^{\alpha} + (\mathscr{C}^{*})^{\beta})\min\{\mu(f), \mu(t, f)\}(s)ds.$$

Next,

$$\begin{aligned}\int_0^t (\mathscr{C}^{\alpha}\min\{\mu(f), \mu(t, f)\})(s)ds &= \mu(t, f)\int_0^t (\mathscr{C}^{\alpha}\chi_{(0,t)})(s)ds\\ &= t\mu(t, f)\int_0^1 (\mathscr{C}^{\alpha}\chi_{(0,1)})(u)du\end{aligned}$$

$$= t\mu(t, f) \cdot (\mathscr{C}^{\alpha+1}\chi_{(0,1)})(1)$$
$$= c_\alpha t\mu(t, f)$$

and

$$\int_0^t ((\mathscr{C}^*)^\beta \min\{\mu(f), \mu(t, f)\})(s)ds$$
$$= \frac{1}{\Gamma(\beta)} \int_0^t \left(\int_s^\infty \min\{\mu(u, f), \mu(t, f)\} \log^{\beta-1}\left(\frac{u}{s}\right) \frac{du}{u} \right) ds$$
$$= t\mu(t, f) + \frac{1}{\Gamma(\beta)} \int_t^\infty \mu(u, f) \left(\int_0^{\frac{t}{u}} \log^{\beta-1}\left(\frac{1}{s}\right) ds \right) du.$$

Using the inequality

$$\int_0^v \log^{\beta-1}\left(\frac{1}{s}\right) ds \le c_\beta v \log^{\beta-1}\left(\frac{1}{v}\right), \quad 0 < v < \frac{1}{2},$$

we estimate

$$\int_0^t ((\mathscr{C}^*)^\beta \min\{\mu(f), \mu(t, f)\})(s)ds$$
$$\le 2t\mu(t, f) + \frac{c_\beta t}{\Gamma(\beta)} \int_{2t}^\infty \mu(u, f) \log^{\beta-1}\left(\frac{u}{t}\right) \frac{du}{u}$$
$$2t\mu(t, f) + c_\beta t(\mathscr{C}^*)^\beta \mu(f)(t).$$

Combining these estimates with (3.5), we obtain

$$\mu(t, T(f_2)) \le c_{\alpha,\beta} \Big((c_\alpha + 2)\mu(f) + c_\beta (\mathscr{C}^*)^\beta \mu(f) \Big)(t). \tag{3.6}$$

Combining (3.4), (3.6) and the last inequality, we have

$$\mu(2t, T(f)) \le \mu(t, T(f_1)) + \mu(t, T(f_2))$$
$$\le \Big(2\mathscr{C}^\alpha \mu(f) + c_{\alpha,\beta}(c_\alpha + 2)\mu(f) + c_{\alpha,\beta} c_\beta (\mathscr{C}^*)^\beta \mu(f) \Big)(t)$$
$$\le \max\{2 + c_{\alpha,\beta}(c_\alpha + 2), c_{\alpha,\beta} c_\beta\} \Big(\mathscr{C}^\alpha \mu(f) + (\mathscr{C}^*)^\beta \mu(f) \Big)(t).$$

Thus, by Lemma 3.7, we have

$$\mu(2t, T(f)) \le c^{(1)}_{\alpha,\beta} \max\{2 + c_{\alpha,\beta}(c_\alpha + 2), c_{\alpha,\beta} c_\beta\} \Big(\mathscr{C}^\alpha \mu(f) + (\mathscr{C}^*)^\beta \mu(f) \Big)(2t).$$

Since $t \in (0, \nu(\mathrm{supp}(f)))$ is arbitrary, the desired assertion follows. □

We are now ready to prove the Theorem 3.1 (ii).

Proof of Theorem 3.1(ii) Take $f \in L_0(\Omega)$ such that $\mu(f) \in \mathrm{dom}(\mathscr{C}^\alpha) \cap \mathrm{dom}((\mathscr{C}^*)^\beta)$. If $\nu(\mathrm{supp})(f) = \infty$, then the assertion follows directly from Lemma 3.9. Suppose now $\nu(\mathrm{supp}(f)) < \infty$. Fix $n \in \mathbb{N}$ and a set $A \subset \mathrm{supp}(f)^c$ such that $\nu(A) = n$. Take $t \in (0, 2\nu(\mathrm{supp}(f)) + 2n)$. Then $t \in (0, 2\nu(\mathrm{supp}(f + \epsilon\chi_A)))$. Since $L_2(\Omega, \mathcal{F}, \nu) \subset L_{2,\infty}(\Omega, \mathcal{F}, \nu)$, it follows that

$$\mu(t, T(\chi_A)) = t^{-\frac{1}{2}} \cdot t^{\frac{1}{2}} \mu(t, T(\chi_A)) \le t^{-\frac{1}{2}} \|T(\chi_A)\|_{L_2(\Omega,\mathcal{F},\nu)} \le 2^{\min\{\alpha,\beta\}} n^{\frac{1}{2}} t^{-\frac{1}{2}}.$$

Hence,

$$\begin{aligned}\mu(2t, Tf) &\le \mu(t, T(f + \epsilon\chi_A)) + \epsilon\mu(t, T(\chi_A)) \\ &\le \mu(t, T(f + \epsilon\chi_A)) + \epsilon 2^{\min\{\alpha,\beta\}} n^{\frac{1}{2}} t^{-\frac{1}{2}}.\end{aligned}$$

Applying Lemma 3.9 to the function $f + \epsilon\chi_A$, we have

$$\begin{aligned}\mu(2t, Tf) &\le \mu(t, T(f + \epsilon\chi_A)) + \epsilon 2^{\min\{\alpha,\beta\}} n^{\frac{1}{2}} t^{-\frac{1}{2}} \\ &\le c'_{\alpha,\beta}(\mathscr{C}^\alpha + (\mathscr{C}^*)^\beta)\mu(f + \epsilon\chi_A)(t) + \epsilon 2^{\min\{\alpha,\beta\}} n^{\frac{1}{2}} t^{-\frac{1}{2}} \\ &\le c'_{\alpha,\beta}(\mathscr{C}^\alpha + (\mathscr{C}^*)^\beta)(\sigma_2\mu(f) + \epsilon\chi_{(0,2n)})(t) + \epsilon 2^{\min\{\alpha,\beta\}} n^{\frac{1}{2}} t^{-\frac{1}{2}} \\ &= c'_{\alpha,\beta}(\mathscr{C}^\alpha + (\mathscr{C}^*)^\beta)\mu(f)\left(\frac{t}{2}\right) + c'_{\alpha,\beta}\epsilon\left((\mathscr{C}^\alpha + (\mathscr{C}^*)^\beta)\chi_{(0,1)}\right)\left(\frac{t}{2n}\right) \\ &\quad + \epsilon 2^{\min\{\alpha,\beta\}} n^{\frac{1}{2}} t^{-\frac{1}{2}},\end{aligned}$$

where we used the fact that the dilation operator commutes with both $\mathscr{C}^\alpha$ and $(\mathscr{C}^*)^\beta$. Since $\epsilon > 0$ is arbitrary, it follows that

$$\mu(2t, Tf) \le c'_{\alpha,\beta}(\mathscr{C}^\alpha + (\mathscr{C}^*)^\beta)\mu(f)\left(\frac{t}{2}\right)$$

for every $t \in (0, 2\nu(\mathrm{supp}(f)) + 2n)$. Since $n \in \mathbb{N}$ is arbitrary, it follows that

$$\mu(2t, Tf) \le c'_{\alpha,\beta}(\mathscr{C}^\alpha + (\mathscr{C}^*)^\beta)\mu(f)\left(\frac{t}{2}\right), \quad t > 0,$$

which, together with Lemma 3.7, gives us

$$\mu(2t, Tf) \le c^{(1)}_{\alpha,\beta} c'_{\alpha,\beta}(\mathscr{C}^\alpha + (\mathscr{C}^*)^\beta)\mu(f)(2t), \quad t > 0.$$

This completes the proof. □

4 Basic estimate associated with Orlicz function

In this section, we study a basic estimate associated with Orlicz function featuring in Theorem 4.1 below. This theorem will help us to show that the commutator with a Calderón–Zygmund integral operator $[M_\phi, T]$ (resp. Marcinkiewicz multiplier T_{m}) satisfies the weak Cesàro property in next two sections.

In this section, we always let M be an Orlicz function, and let ψ_M be a concave majorant of the fundamental function on $L_M(0,1)$, that is,

$$\frac{1}{M^{-1}(\frac{1}{u})} \leq \psi_M(u), \quad u \in (0,1). \tag{4.1}$$

We here refer the reader to [6, p. 65] for the definition of fundamental function of a symmetric space.

Theorem 4.1 *If $\{S_k\}_{k\geq 1}$ are pairwise disjoint sub-intervals of $(0,t)$ for some $t \in (0,\infty)$, then*

$$\sum_{k\geq 1} m(S_k)\|f\|_{L_M(S_k,\frac{m}{m(S_k)})} \leq \int_0^t \mu(s,f)\psi_M'\left(\frac{s}{t}\right)ds, \quad f \in L_\infty(0,t).$$

The proof of the following lemma was suggested to us by the anonymous referee (our initial proof was excessively complicated).

Lemma 4.2 *Let $\mathfrak{S}$ be the collection of sequences of pairwise disjoint subsets of $(0,1)$. Set*

$$\psi_{\mathfrak{S},M}(s) = \sup_{\{S_k\}_{k\geq 1}\in\mathfrak{S}} \sum_{k\geq 1} m(S_k)\|\chi_{(0,s)}\|_{L_M(S_k,\frac{m}{m(S_k)})}, \quad s \in (0,1).$$

We have

$$\psi_{\mathfrak{S},M}(s) \leq \psi_M(s), \quad s \in (0,1).$$

Proof Without loss of generality, let $\bigcup_{k\geq 1} S_k = (0,1)$. Indeed, otherwise we set $S_0 = (0,1)\setminus\bigcup_{k\geq 1} S_k$ and consider the sequence $\{S_k\}_{k\geq 0}$ instead of the sequence $\{S_k\}_{k\geq 1}$.

Clearly, for each $s \in (0,1)$, we have

$$\psi_{\mathfrak{S},M}(s) = \sup_{\{S_k\}_{k\geq 1}\in\mathfrak{S}} \sum_{k\geq 1} m(S_k)\frac{1}{M^{-1}(\frac{m(S_k)}{m((0,s)\cap S_k)})},$$

which, together with (4.1), implies

$$\psi_{\mathfrak{S},M}(s) \leq \sup_{\{S_k\}_{k\geq 1}\in\mathfrak{S}} \sum_{k\geq 1} m(S_k)\psi_M\left(\frac{m((0,s)\cap S_k)}{m(S_k)}\right), \quad s \in (0,1).$$

Denote

$$\alpha_k = m(S_k), \quad \theta_k = \frac{m((0,s)\cap S_k)}{m(S_k)}, \quad k \geq 1.$$

We have $\alpha_k \geq 0$, $\theta_k \in [0,1]$ and

$$\sum_{k\geq 1}\alpha_k = 1, \quad \sum_{k\geq 1}\alpha_k\theta_k = s.$$

By using concavity of ψ_M,

$$\sum_{k\geq 1} m(S_k)\psi_M\left(\frac{m((0,s)\cap S_k)}{m(S_k)}\right) = \sum_{k\geq 1}\alpha_k\psi_M(\theta_k) \leq \psi_M\left(\sum_{k\geq 1}\alpha_k\theta_k\right) = \psi_M(s).$$

This completes the proof. □

We are now ready to prove Theorem 4.1.

Proof of Theorem 4.1 Both assumption and assertion are dilation invariant. Hence, we may assume without loss of generality that $t = 1$.

Let $\mathfrak{S}$ be the collection of sequence of pairwise disjoint subsets of $(0,1)$. Consider the symmetric norm $\|\cdot\|_E$ on $L_\infty(0,1)$ defined by the formula

$$\|f\|_E = \sup_{\{S_k\}_{k\geq 1}\in\mathfrak{S}} \sum_{k\geq 1} m(S_k)\|f\|_{L_M(S_k,\frac{m}{m(S_k)})}, \quad f \in L_\infty(0,1).$$

Clearly, the fundamental function of $\|\cdot\|_E$ is $\psi_{\mathfrak{S},M}$ defined in Lemma 4.2. Hence,

$$\|f\|_E \leq \|f\|_{\Lambda_{\psi_{\mathfrak{S},M}}}, \quad f \in L_\infty(0,1).$$

Using Lemma 4.2, we write

$$\|f\|_E \leq \int_0^1 \mu(s,f)d\psi_M(s) = \int_0^1 \mu(s,f)\psi_M'(s)ds.$$

This completes the proof. □

5 Proof of Theorem 1.2

In this section, we prove Theorem 1.2. For the upper estimate, with the help of Theorem 3.1, we show in Theorem 5.1 that the commutator $[M_\phi, T]$ satisfies the weak Cesàro property WCP(2) (see Definition 1.7). In Sect. 5.2, we show that the upper estimate in Theorem 1.2 is optimal.

5.1 Upper estimate

The bulk of this subsection is occupied with the proof of the following result.

Theorem 5.1 *Let $\phi \in \mathrm{BMO}(\mathbb{R}^d)$ be such that $\|\phi\|_{\mathrm{BMO}(\mathbb{R}^d)} = 1$. Let T be a Calderón–Zygmund integral operator as in Assumption 1.1. The commutator $[M_\phi, T]$ satisfies* WCP(2)*, that is, for every $0 \le f \in (L_1 \cap L_\infty)(\mathbb{R}^d)$ supported on a set of finite measure and for every $\lambda \ge \mathscr{C}\mu(f)(m(\mathrm{supp}(f)))$, there exists a positive constant $c_{d,T}$ such that*

$$\lambda m\Big(\Big\{y \in \mathbb{R}^d : \big|[M_\phi, T]f\big|(y) > 3\lambda\Big\}\Big) \le c_{d,T} \int_0^{m(\mathrm{supp}(f))} \mathscr{C}\mu(f)(s)ds.$$

To prove Theorem 5.1, we need several lemmas. Set $\Phi(t) = t\log(1+t) + t$ for $t \ge 0$, and set $\Phi^*(t) = e^t - t - 1$ for $t \ge 0$. It is easy to see that

$$ts \le \Phi(t) + \Phi^*(s), \quad t, s \ge 0.$$

It follows from [38, Theorem 4.7.5] that, for each pair $(f, g) \in L_\Phi(0,1) \times L_{\Phi^*}(0,1)$, we have

$$\int_0^1 |fg|(t)dt \le 2\|f\|_{L_\Phi(0,1)}\|g\|_{L_{\Phi^*}(0,1)}.$$

Consequently,

$$\|fg\|_{L_1(Q)} = m(Q)\|fg\|_{L_1(Q,\frac{m}{m(Q)})} \le 2m(Q)\|f\|_{L_\Phi(Q,\frac{m}{m(Q)})}\|g\|_{L_{\Phi^*}(Q,\frac{m}{m(Q)})} \quad (5.1)$$

for every cube Q in $\mathbb{R}^d$. During the proof, we frequently use the fact that for each $f \in (L_1 \cap L_\infty)(\mathbb{R}^d)$,

$$\|f\|_{L_1(\mathbb{R}^d)} \le m(\mathrm{supp}(f)) \cdot \mathscr{C}^\alpha\mu(f)(m(\mathrm{supp}(f)), \quad \alpha \ge 1. \quad (5.2)$$

Lemma 5.2 *Let $0 \le f \in (L_1 \cap L_\infty)(\mathbb{R}^d)$ be supported on a set of finite measure, and let $\phi \in \mathrm{BMO}(\mathbb{R}^d)$ with $\|\phi\|_{\mathrm{BMO}(\mathbb{R}^d)} = 1$. Let $\lambda \ge \mathscr{C}\mu(f)(m(\mathrm{supp}(f)))$ and let $\{Q\}_{Q\in\mathcal{Q}}$ be as in Construction 2.4. We have*

$$\Big\|\sum_{Q\in\mathcal{Q}}(\phi - \phi_Q)(f - f_Q)\chi_Q\Big\|_{L_1(\mathbb{R}^d)} \le 2^{d+8}\int_0^{m(\mathrm{supp}(f))} \mathscr{C}\mu(f)(s)ds.$$

Proof Let us begin with

$$\Big\|\sum_{Q\in\mathcal{Q}}(\phi - \phi_Q)(f - f_Q)\chi_Q\Big\|_{L_1(\mathbb{R}^d)}$$

$$\leq \sum_{Q\in\mathcal{Q}} \|(\phi-\phi_Q)(f-f_Q)\|_{L_1(Q)}$$
$$\leq \sum_{Q\in\mathcal{Q}} 2m(Q)\|f-f_Q\|_{L_\Phi(Q,\frac{m}{m(Q)})}\|\phi-\phi_Q\|_{L_{\Phi^*}(Q,\frac{m}{m(Q)})}$$
$$\leq \sum_{Q\in\mathcal{Q}} 2^{d+3}e^2 m(Q)\|f-f_Q\|_{L_\Phi(Q,\frac{m}{m(Q)})},$$

where the last two inequalities are due to (5.1) and Lemma 2.3. Thus,

$$\Big\|\sum_{Q\in\mathcal{Q}}(\phi-\phi_Q)(f-f_Q)\chi_Q\Big\|_{L_1(\mathbb{R}^d)} \leq 2^{d+3}e^2\sum_{Q\in\mathcal{Q}} m(Q)\|f\|_{L_\Phi(Q,\frac{m}{m(Q)})} + 2^{d+3}e^2\sum_{Q\in\mathcal{Q}} f_Q m(Q). \tag{5.3}$$

According to Construction 2.4, we know that the union of the disjoint dyadic cubes $\{Q\}_{Q\in\mathcal{Q}}$ is Ω_λ. By [6], there exists a measure-preserving map from Ω_λ to the interval $[0, m(\Omega_\lambda)]$. Under this map, $\{Q\}_{Q\in\mathcal{Q}}$ becomes a collection of pairwise disjoint measurable subsets in $[0, m(\Omega_\lambda)]$. Recall from Lemma A.1 that

$$\frac{1}{\Phi^{-1}(\frac{1}{s})} \leq 2s\log\left(\frac{e}{s}\right), \quad 0<s<1.$$

Now, applying Theorem 4.1 and taking into account that composition with a measure-preserving map which does not change μ, we infer

$$\sum_{Q\in\mathcal{Q}} m(Q)\|f\|_{L_\Phi(Q,\frac{m}{m(Q)})} \leq 2\int_0^{m(\Omega_\lambda)} \mathscr{C}\mu(f)(s)ds.$$

Note that the assumption on λ, i.e., $\lambda \geq \mathscr{C}\mu(f)(m(\mathrm{supp}(f)))$, implies

$$\lambda^{-1} \leq m(\mathrm{supp}(f))\|f\|^{-1}_{L_1(\mathbb{R}^d)}. \tag{5.4}$$

Thus, using Lemma 2.5, we have

$$m(\Omega_\lambda) \leq \lambda^{-1}\|f\|_{L_1(\mathbb{R}^d)} \leq m(\mathrm{supp}(f)).$$

Hence,

$$\sum_{Q\in\mathcal{Q}} m(Q)\|f\|_{L_\Phi(Q,\frac{m}{m(Q)})} \leq 2\int_0^{m(\mathrm{supp}(f))} \mathscr{C}\mu(f)(s)ds. \tag{5.5}$$

On the other hand, we have

$$\sum_{Q\in\mathcal{Q}} f_Q m(Q) = \int_{\Omega_\lambda} f(y)dy \le \|f\|_{L_1(\mathbb{R}^d)} \le \int_0^{m(\mathrm{supp}(f))} \mathscr{C}\mu(f)(s)ds. \quad (5.6)$$

Combining (5.3), (5.5) and (5.6), we obtain

$$\Big\| \sum_{Q\in\mathcal{Q}} (\phi - \phi_Q)(f - f_Q)\chi_Q \Big\|_{L_1(\mathbb{R}^d)} \le 3\cdot 2^{d+3}e^2 \cdot \int_0^{m(\mathrm{supp}(f))} \mathscr{C}\mu(f)(s)ds.$$

This gives the assertion. □

We need more estimates related to BMO functions. Some of them should be known for experts, and we include the proof for the sake of completeness. In what follows, for each cube $Q \subset \mathbb{R}^d$, c_Q denotes the center of Q, and rQ (here $r \in (0,\infty)$) is the dilation of Q with respect to its center.

Lemma 5.3 *Let $\phi \in \mathrm{BMO}(\mathbb{R}^d)$. For every cube $Q \subset \mathbb{R}^d$, we have*

$$|\phi_{2Q} - \phi_Q| \le 2^d \|\phi\|_{\mathrm{BMO}(\mathbb{R}^d)}.$$

Proof We have

$$\begin{aligned} |\phi_Q - \phi_{2Q}| &= \Big|\frac{1}{m(Q)}\int_Q (\phi(x) - \phi_{2Q})dx\Big| \le \frac{1}{m(Q)}\int_Q |\phi(x) - \phi_{2Q}|dx \\ &\le \frac{1}{m(Q)}\int_{2Q} |\phi(x) - \phi_{2Q}|dx \le 2^d\|\phi\|_{\mathrm{BMO}(\mathbb{R}^d)}. \end{aligned}$$

□

Lemma 5.4 *Let $\phi \in \mathrm{BMO}(\mathbb{R}^d)$ with $\|\phi\|_{\mathrm{BMO}(\mathbb{R}^d)} = 1$. For every cube $Q \subset \mathbb{R}^d$, we have*

$$\int_{\mathbb{R}^d\setminus 2Q} \frac{|\phi(y) - \phi_Q|}{|y - c_Q|^{d+1}}dy \le \frac{2^{4d+4}}{\ell(Q)},$$

where $\ell(Q)$ denotes the length side of Q.

Proof Recall that c_Q denotes the center of Q. Write

$$\int_{\mathbb{R}^d\setminus 2Q} \frac{|\phi(y) - \phi_Q|}{|y - c_Q|^{d+1}}dy = \sum_{k\ge 1}\int_{2^{k+1}Q\setminus 2^kQ} \frac{|\phi(y) - \phi_Q|}{|y - c_Q|^{d+1}}dy.$$

Note that for $y \in 2^{k+1}Q\setminus 2^kQ$,

$$|y - c_Q| \ge \frac{2^k - 1}{2}\ell(Q) \ge 2^{k-2}\ell(Q).$$

Hence, for each $k \geq 1$, we have

$$\begin{aligned}\int_{2^{k+1}Q\setminus 2^k Q} \frac{|\phi(y)-\phi_Q|}{|y-c_Q|^{d+1}}dy &\leq \frac{1}{(2^{k-2}\ell(Q))^{d+1}}\int_{2^{k+1}Q\setminus 2^k Q}|\phi(y)-\phi_Q|dy\\ &\leq \frac{1}{(2^{k-2}\ell(Q))^{d+1}}\int_{2^{k+1}Q}|\phi(y)-\phi_Q|dy\\ &= \frac{8^{d+1}}{2^{k+1}\ell(Q)}\cdot\frac{1}{m(2^{k+1}Q)}\int_{2^{k+1}Q}|\phi(y)-\phi_Q|dy.\end{aligned}$$

By the triangle inequality,

$$\int_{2^{k+1}Q}|\phi(y)-\phi_Q|dy \leq \int_{2^{k+1}Q}|\phi(y)-\phi_{2^{k+1}Q}|dy + m(2^{k+1}Q)|\phi_{2^{k+1}Q}-\phi_Q|.$$

By the definition of the BMO norm, we have

$$\frac{1}{m(2^{k+1}Q)}\int_{2^{k+1}Q}|\phi(y)-\phi_{2^{k+1}Q}|dy \leq \|\phi\|_{\mathrm{BMO}(\mathbb{R}^d)} = 1.$$

Clearly, by Lemma 5.3,

$$|\phi_{2^{k+1}Q}-\phi_Q| \leq \sum_{l=0}^{k}|\phi_{2^{l+1}Q}-\phi_{2^lQ}| \leq 2^d\sum_{l=0}^{k}\|\phi\|_{\mathrm{BMO}(\mathbb{R}^d)} = 2^d(k+1).$$

Therefore,

$$\frac{1}{m(2^{k+1}Q)}\int_{2^{k+1}Q}|\phi(y)-\phi_Q|dy \leq (1+2^d(k+1)), \quad k\geq 1.$$

Combining these inequalities, we obtain

$$\int_{\mathbb{R}^d\setminus 2Q}\frac{|\phi(y)-\phi_Q|}{|y-c_Q|^{d+1}}dy \leq \sum_{k\geq 1}\frac{8^{d+1}}{2^{k+1}\ell(Q)}(1+2^d(k+1)) \leq \frac{2^{4d+4}}{\ell(Q)}.$$

This is the assertion. □

Lemma 5.5 *Let $0 \leq f \in (L_1\cap L_\infty)(\mathbb{R}^d)$ and let $\phi \in \mathrm{BMO}(\mathbb{R}^d)$ with $\|\phi\|_{\mathrm{BMO}(\mathbb{R}^d)} = 1$. Let $\lambda > 0$ and let $\{Q\}_{Q\in\mathcal{Q}}$ be as in Construction 2.4. Let T be a Calderón–Zygmund integral operator as in Assumption 1.1. There exists a positive constant c_T such that*

$$\Big\|\sum_{Q\in\mathcal{Q}}(\phi-\phi_Q)\cdot T((f-f_Q)\chi_Q)\Big\|_{L_1(\mathbb{R}^d\setminus\cup_{Q\in\mathcal{Q}}(2d+1)Q)} \leq c_T\|f\|_{L_1(\mathbb{R}^d)}.$$

Proof Since $(f - f_Q)\chi_Q$ is mean zero, it follows that for each $y \in \mathbb{R}^d \setminus \cup_{Q\in\mathcal{Q}} (2d+1)Q$,

$$\begin{aligned} T((f-f_Q)\chi_Q)(y) &= \int_Q K(y,x)(f-f_Q)(x)dx \\ &= \int_Q (K(y,x) - K(y,c_Q))(f-f_Q)(x)dx. \end{aligned}$$

Taking into account that

$$|y - c_Q| \geq 2|x - c_Q|, \quad x \in Q, \quad y \in (2d+1)Q,$$

we conclude from Assumption 1.1 (ii) that

$$|K(y,x) - K(y,c_Q)| \leq c_T \frac{|x-c_Q|}{|y-c_Q|^{d+1}} \leq c_T \frac{d\ell(Q)}{|y-c_Q|^{d+1}}, \quad x \in Q, y \notin (2d+1)Q.$$

Thus, for each $y \notin (2d+1)Q$, we have

$$\begin{aligned} |T((f-f_Q)\chi_Q)(y)| &\leq c_T \frac{d\ell(Q)}{|y-c_Q|^{d+1}} \cdot \int_Q |f(x) - f_Q|dx \\ &\leq 2c_T \frac{d\ell(Q)}{|y-c_Q|^{d+1}} \cdot \int_Q f(x)dx. \end{aligned}$$

Consequently, by Lemma 5.4,

$$\begin{aligned} &\left\|(\phi - \phi_Q)\cdot T((f-f_Q)\chi_Q)\right\|_{L_1(\mathbb{R}^d\setminus\cup_{Q\in\mathcal{Q}}(2d+1)Q)} \\ &\leq 2dc_T\ell(Q)\cdot \int_{\mathbb{R}^d\setminus(2d+1)Q} \frac{|\phi(y)-\phi_Q|}{|y-c_Q|^{d+1}}dy \cdot \int_Q f(x)dx \\ &\leq 2^{4d+5}dc_T \cdot \int_Q f(x)dx. \end{aligned}$$

Finally, we conclude the above estimates that

$$\left\| \sum_{Q\in\mathcal{Q}} (\phi - \phi_Q)\cdot T((f-f_Q)\chi_Q)\right\|_{L_1(\mathbb{R}^d\setminus\cup_{Q\in\mathcal{Q}}(2d+1)Q)} \leq 2^{4d+5}dc_T\|f\|_{L_1(\mathbb{R}^d)}.$$

The proof is complete. □

The following result is taken from [17, 24].

Lemma 5.6 [24, Theorem 1.1], [17, Theorem 1.3] *Let T be a Calderón–Zygmund integral operator with kernel satisfying Assumption 1.1, and let $\phi \in \mathrm{BMO}(\mathbb{R}^d)$. The*

commutator $[M_\phi, T]$ is bounded on $L_2(\mathbb{R}^d)$, that is there is a constant $c_T^{(1)}$ depending on d and T such that

$$\|[M_\phi, T]f\|_{L_2} \le c_T^{(1)}\|\phi\|_{\mathrm{BMO}(\mathbb{R}^d)}\|f\|_{L_2(\mathbb{R}^d)}, \quad f \in L_2(\mathbb{R}^d).$$

Now we are ready to prove Theorem 5.1.

Proof of Theorem 5.1 Let $0 \le f \in (L_1 \cap L_\infty)(\mathbb{R}^d)$ supported on a set of finite measure. Consider the Calderón–Zygmund decomposition as in Construction 2.4 for f at level $\lambda > 0$. We write

$$[M_\phi, T]f = [M_\phi, T]g + [M_\phi, T]b.$$

We have

$$\begin{aligned} & m\left(\left\{y \in \mathbb{R}^d : \left|[M_\phi, T]f\right|(y) > 3\lambda\right\}\right) \\ & \le m\left(\bigcup_{Q\in\mathcal{Q}} (2d+1)Q\right) + m\left(\left\{y \in \mathbb{R}^d : \left|[M_\phi, T]g\right|(y) > \lambda\right\}\right) \\ & \quad + m\left(\left\{y \in \mathbb{R}^d \setminus \bigcup_{Q\in\mathcal{Q}} (2d+1)Q : \left|[M_\phi, T]b\right|(y) > 2\lambda\right\}\right) \\ & = \mathrm{I} + \mathrm{II} + \mathrm{III}. \end{aligned}$$

It suffices to estimate each term on the right hand side separately. By Construction 2.4 and Lemma 2.5, we have

$$\begin{aligned} \mathrm{I} \le \sum_{Q\in\mathcal{Q}} m((2d+1)Q) &= (2d+1)^d \sum_{Q\in\mathcal{Q}} m(Q) \\ &= (2d+1)^d m(\Omega_\lambda) \le (2d+1)^d \lambda^{-1}\|f\|_{L_1(\mathbb{R}^d)}. \end{aligned}$$

By the Chebyshev inequality, Lemma 2.5 and 5.6, we have

$$\mathrm{II} \le \frac{1}{\lambda^2}\left\|[M_\phi, T]g\right\|^2_{L_2(\mathbb{R}^d)} \le (c_T^{(1)})^2 \frac{1}{\lambda^2}\|g\|^2_{L_2(\mathbb{R}^d)} \le 2^d (c_T^{(1)})^2 \frac{1}{\lambda^2}\lambda\|f\|_{L_1(\mathbb{R}^d)}.$$

It remains to estimate III. According to the definition of b as in Construction 2.4, we write

$$\begin{aligned} [M_\phi, T]b &= \sum_{Q\in\mathcal{Q}} [M_\phi, T]((f - f_Q)\chi_Q) = \sum_{Q\in\mathcal{Q}} [M_{\phi-\phi_Q}, T]((f - f_Q)\chi_Q) \\ &= \sum_{Q\in\mathcal{Q}} (\phi - \phi_Q)\cdot T((f - f_Q)\chi_Q) - T\left(\sum_{Q\in\mathcal{Q}} (\phi - \phi_Q)(f - f_Q)\chi_Q\right). \end{aligned}$$

Note that T is of weak type $(1,1)$; see Theorem A. We have

$$\begin{aligned}
\mathrm{III} &\le m\left(\left\{y\in\mathbb{R}^d : \left|T\left(\sum_{Q\in\mathcal{Q}}(\phi-\phi_Q)(f-f_Q)\chi_Q\right)\right|(y)>\lambda\right\}\right)\\
&\quad + m\left(\left\{y\in\mathbb{R}^d\setminus\bigcup_{Q\in\mathcal{Q}}(2d+1)Q : \left|\sum_{Q\in\mathcal{Q}}(\phi-\phi_Q)\cdot T((f-f_Q)\chi_Q)\right|(y)>\lambda\right\}\right)\\
&\le \lambda^{-1}\left\|T\left(\sum_{Q\in\mathcal{Q}}(\phi-\phi_Q)(f-f_Q)\chi_Q\right)\right\|_{L_{1,\infty}(\mathbb{R}^d)}\\
&\quad + \lambda^{-1}\left\|\sum_{Q\in\mathcal{Q}}(\phi-\phi_Q)\cdot T((f-f_Q)\chi_Q)\right\|_{L_1(\mathbb{R}^d\setminus\bigcup_{Q\in A}(2d+1)Q)}\\
&\le \lambda^{-1}c_T\left\|\sum_{Q\in\mathcal{Q}}(\phi-\phi_Q)(f-f_Q)\chi_Q\right\|_{L_1(\mathbb{R}^d)}\\
&\quad + \lambda^{-1}\left\|\sum_{Q\in\mathcal{Q}}(\phi-\phi_Q)\cdot T((f-f_Q)\chi_Q)\right\|_{L_1(\mathbb{R}^d\setminus\bigcup_{Q\in\mathcal{Q}}(2d+1)Q)}.
\end{aligned}$$

By Lemmas 5.2 and 5.5, we have

$$\begin{aligned}
&\lambda m\left(\left\{y\in\mathbb{R}^d\setminus\bigcup_{Q\in A}(2d+1)Q : \left|[M_\phi,T]b\right|(y)>2\lambda\right\}\right)\\
&\quad\le 2^{d+8}c_T\cdot\int_0^{m(\mathrm{supp}(f))}(\mathscr{C}\mu(f))(s)ds + c_T\|f\|_{L_1(\mathbb{R}^d)}.
\end{aligned}$$

Combining these inequalities and (5.2), we arrive at

$$\lambda m\left(\left\{y\in\mathbb{R}^d : \left|[M_\phi,T]f\right|(y)>3\lambda\right\}\right)\le c_{d,T}\int_0^{m(\mathrm{supp}(f))}(\mathscr{C}\mu(f))(s)ds.$$

This completes the proof. □

We are going to prove Theorem 1.2. However, we still need to know that the behavior of the commutator $[M_\phi,T]$ acting on $L_p(\mathbb{R}^d)$, $1<p\le 2$. To this end, we establish the following interpolation lemma, which also shows that the weak Cesàro property introduced in Definition 1.7 is also a suitable endpoint estimate for interpolation purpose.

Lemma 5.7 *Let $\alpha > 0$, and let $\mathcal{Z} : L_2(\mathbb{R}^d) \to L_2(\mathbb{R}^d)$ be a linear contraction satisfying the property* WCP(α). *We have*

$$\|\mathcal{Z}\|_{L_p(\mathbb{R}^d)\to L_p(\mathbb{R}^d)} \le 32(2p')^\alpha, \quad 1 < p \le 2.$$

Proof Suppose first $1 < p \le \frac{3}{2}$.

Let $f \in L_p(\mathbb{R}^d)$ whose support has finite measure. Fix $t > 0$ and decompose $f = f_1 + f_2$ so that

$$\mu(u, f_1) = \mu(u, f)\chi_{(0,t)}(u), \quad \mu(u, f_2) = \mu(u + t, f), \quad u > 0.$$

Note that $m(\mathrm{supp}(f_1)) = t$. By Lemma 3.8, we have

$$\mu(t, \mathcal{Z}(f_1)) \le 2\mathscr{C}^\alpha \mu(f_1)(t) \le 2\mathscr{C}^\alpha \mu(f)(t).$$

On the other hand, we have

$$t\mu^2(t, \mathcal{Z}f_2) \le \|\mathcal{Z}f_2\|_2^2 \le \|f_2\|_2^2 = \int_t^\infty \mu^2(s, f)ds.$$

Hence,

$$\begin{aligned}\mu(2t, \mathcal{Z}f) &\le \mu(t, \mathcal{Z}(f_1)) + \mu(t, \mathcal{Z}(f_2)) \\ &\le 2\mathscr{C}^\alpha \mu(f)(t) + \left(\frac{1}{t}\int_t^\infty \mu^2(s, f)ds\right)^{\frac{1}{2}}, \quad t > 0. \end{aligned} \tag{5.7}$$

Denote for brevity the operator $\mathcal{V} : L_2(0, \infty) \to L_2(0, \infty)$ by setting

$$(\mathcal{V}h)(t) = \left(\frac{1}{t}\int_t^\infty \mu^2(s, h)ds\right)^{\frac{1}{2}}, \quad t > 0.$$

By (5.7), we have

$$\mu(\mathcal{Z}f) \le 2\sigma_2\mathscr{C}^\alpha \mu(f) + 2\sigma_2\mathcal{V}\mu(f).$$

Consequently,

$$\begin{aligned}\|\mathcal{Z}f\|_{L_p(\mathbb{R}^d)} &\le 2^{1+\frac{1}{p}}\|\mathscr{C}^\alpha \mu(f)\|_{L_p(0,\infty)} + 2^{1+\frac{1}{p}}\|\mathcal{V}\mu(f)\|_{L_p(0,\infty)} \\ &\le 2^{1+\frac{1}{p}}\|\mathscr{C}^\alpha\|_{L_p(0,\infty)\to L_p(0,\infty)}\|f\|_{L_p(\mathbb{R}^d)} \\ &\quad + 2^{1+\frac{1}{p}}\|\mathcal{V}\|_{L_p(0,\infty)\to L_p(0,\infty)}\|f\|_{L_p(\mathbb{R}^d)}.\end{aligned}$$

Taking into account that

$$\|\mathscr{C}\|_{L_p(0,\infty)\to L_p(0,\infty)} \overset{(2.8)}{\leq} (p')^\alpha, \quad 1 < p < \infty,$$

and

$$\|\mathcal{V}\|_{L_p(0,\infty)\to L_p(0,\infty)} \leq (1 - 2^{\frac{p}{2}-1})^{\frac{1}{p}}, \quad 0 < p < 2,$$

we infer

$$\|\mathcal{Z}f\|_{L_p(\mathbb{R}^d)} \leq (4(p')^\alpha + 28)\|f\|_{L_p(\mathbb{R}^d)} \leq 32(p')^\alpha \|f\|_{L_p(\mathbb{R}^d)}, \quad 1 < p \leq \frac{3}{2}.$$

It immediately follows that exactly the same inequality holds for every $f \in L_p(\mathbb{R}^d)$. Hence,

$$\|\mathcal{Z}\|_{L_p(\mathbb{R}^d)\to L_p(\mathbb{R}^d)} \leq 32(p')^\alpha, \quad 1 < p \leq \frac{3}{2}.$$

In particular,

$$\|\mathcal{Z}\|_{L_{\frac{3}{2}}(\mathbb{R}^d)\to L_{\frac{3}{2}}(\mathbb{R}^d)} \leq 32 \cdot 3^\alpha.$$

Since also $\|\mathcal{Z}\|_{L_2(\mathbb{R}^d)\to L_2(\mathbb{R}^d)} \leq 1$, it follows from complex interpolation that

$$\|\mathcal{Z}\|_{L_p(\mathbb{R}^d)\to L_p(\mathbb{R}^d)} \leq 32 \cdot 3^\alpha \leq 32 \cdot (2p')^\alpha, \quad \frac{3}{2} \leq p \leq 2.$$

This completes the proof. □

Now we are ready to prove the upper estimate in Theorem 1.2.

Proof of Theorem 1.2: upper estimate. Without loss of generality, assume that the Calderón–Zygmund operator T is self-adjoint, and $\phi \in \mathrm{BMO}(\mathbb{R}^d)$ is real-valued and such that $\|\phi\|_{\mathrm{BMO}(\mathbb{R}^d)} = 1$. Hence, the commutator $i[M_\phi, T]$ is a self-adjoint operator.

Since Theorem 5.1 states that the commutator $[M_\phi, T]$ satisfies the weak Cesàro property WCP(2), it follows from Lemma 5.7 and duality that

$$\|[M_\phi, T]\|_{L_p(\mathbb{R}^d)\to L_p(\mathbb{R}^d)} \leq \begin{cases} \frac{c_T}{(p-1)^2}, & 1 < p \leq 2 \\ c_T p^2, & 2 \leq p < \infty. \end{cases}$$

The desired assertion now follows from Theorems 3.1 and 5.1. □

5.2 Lower estimate: optimality

We continue to show that our lower estimate is optimal. The example given here is taken from Pérez [37].

Let $d = 1$, $T = H$ be the Hilbert transform, and let $\phi \in \mathrm{BMO}(\mathbb{R})$ defined by setting $\phi(t) = \log(|t|)$. Let f be a positive function supported on $(0, \infty)$ such that $f = \mu(f) \in \mathrm{dom}(\mathscr{C}^2) \cap \mathrm{dom}((\mathscr{C}^*)^2)$. For $t > 0$, we have

$$\begin{aligned}\pi([M_\phi, T]f)(t) &= \int_0^\infty \frac{\log(t) - \log(s)}{t - s} f(s)ds \\ &= \int_0^t \frac{\log(t) - \log(s)}{t - s} f(s)ds + \int_t^\infty \frac{\log(s) - \log(t)}{s - t} f(s)ds \\ &\geq \int_0^t \frac{\log(t) - \log(s)}{t} f(s)ds + \int_t^\infty \frac{\log(s) - \log(t)}{s} f(s)ds \\ &= (\mathscr{C}^2 f + (\mathscr{C}^*)^2 f)(t),\end{aligned}$$

where the last equality can be deduced from the definitions (2.4) and (2.5) directly. Consequently,

$$\mu([M_\phi, H]f) \geq \frac{1}{\pi}(\mathscr{C}^2 + (\mathscr{C}^*)^2)\mu(f).$$

This is the expected optimality estimate.

6 Proofs of Theorems 1.4 and 1.5

6.1 Proof of Theorem 1.4

In this subsection, we first show that the Marcinkiewicz multiplier T_{m} satisfies the property WCP($\frac{3}{2}$) in Proposition 6.6, and then prove Theorem 1.4 via Theorem 3.1. To this end, let us recall Bakas et al.'s recent result [3] concerning Marcinkiewicz multipliers. In what follows, let us denote

$$\Phi(t) = t \log^{\frac{1}{2}}(e + t), \quad t > 0.$$

The following Calderón–Zygmund decomposition associated to an Orlicz function was established in [16] (see Theorem 5.2, Proposition A.1 and Remark A.3 there). It is crucially used in [3] (see Proposition 5.2 there).

Construction 6.1 *Let $0 \leq f \in L_\Phi(\mathbb{R})$ be supported on a set of finite measure. Consider the function*

$$M_\Phi f : x \to \sup_{\substack{Q \in \mathcal{D} \\ x \in Q}} \|f\|_{L_\Phi(Q, \frac{m}{m(Q)})}, \quad x \in \mathbb{R},$$

where $\mathcal{D}$ is the dyadic system introduced in Sect. 2.3. For a given $\lambda > 0$, set

$$\Sigma_\lambda = \Big\{x \in \mathbb{R} : \ (M_\Phi f)(x) > \lambda\Big\}.$$

Then there exists a subset $\mathcal{K}_\lambda \subset \mathcal{D}$ with disjoint dyadic cubes on $\mathbb{R}$ such that

$$\Sigma_\lambda = \bigcup_{Q \in \mathcal{K}_\lambda} Q$$

such that

$$\lambda \le \|f\|_{L_\Phi(Q, \frac{m}{m(Q)})} \le 2\lambda, \quad Q \in \mathcal{K}_\lambda.$$

We also have

$$m(\Sigma_\lambda) \le 3 \cdot 2^6 \int_{\mathbb{R}} \Phi\left(\frac{|f(x)|}{\lambda}\right) dx.$$

Applying Construction 6.1, we have the following estimate.

Theorem 6.2 *Let T_{m} be a Marcinkiewicz multiplier on $\mathbb{R}$. Let $\lambda > 0$ and let Σ_λ be as in Construction 6.1. There exists a positive constant $c_{T_{\mathsf{m}}}$ depending on m such that*

$$m\Big(\Big\{y \in \mathbb{R} : \ |T_{\mathsf{m}} f(y)| > \lambda\Big\}\Big) \le c_{T_{\mathsf{m}}}\Big(m(\Sigma_\lambda) + \lambda^{-1}\|f\|_{L_1(\mathbb{R}^d)}\Big)$$

for each $0 \le f \in L_\Phi(\mathbb{R})$ supported on a set of finite measure.

Remark 6.3 For the proof of the above result, we refer the reader to [3, Section 6]. The authors are indebted to Professor Bakas, who confirmed that a quick inspection of the proofs given in [3, Section 6] yields not only Theorem C (see Introduction), but Theorem 6.2 as well.

For the proof of Theorem 1.4 we need some auxiliary estimates.

Lemma 6.4 *Under Construction 6.1, if $\lambda \ge \mathscr{C}^{\frac{3}{2}}\mu(f)(m(\mathrm{supp}(f)))$, then*

$$m(\Sigma_\lambda) \le 2^{10} m(\mathrm{supp}(f)).$$

Proof For brevity, write $A = \mathrm{supp}(f)$. We have

$$\|f\|_{L_\Phi(A, \frac{m}{m(A)})} = \|\sigma_{\frac{1}{m(A)}}\mu(f)\|_{L_\Phi(0,1)}.$$

By Lemma A.2, the function $t \to 2t\log^{\frac{1}{2}}(\frac{e}{t})$ is a concave majorant for the fundamental function of $L_\Phi(0, 1)$. Hence,

$$\|h\|_{L_\Phi(0,1)} \le 2\|h\|_{\Lambda_\psi(0,1)} \overset{\text{Lemma A.3}}{\le} 2\int_0^1 (\mathscr{C}^{\frac{1}{2}}\mu(h))(t)dt, \quad \forall h \in L_\Phi(0, 1),$$

we obtain

$$\begin{aligned}\|f\|_{L_\Phi(A,\frac{m}{m(A)})} &\le 2\int_0^1 \Big(\mathscr{C}^{\frac{1}{2}}(\sigma_{\frac{1}{m(A)}}\mu(f))\Big)(t)dt\\ &= 2\int_0^1 \Big(\mathscr{C}^{\frac{1}{2}}\mu(f)\Big)(m(A)\cdot t)dt\\ &= \frac{2}{m(A)}\int_0^{m(A)} \Big(\mathscr{C}^{\frac{1}{2}}\mu(f)\Big)(s)ds\\ &= 2\mathscr{C}^{\frac{3}{2}}\mu(f)(m(A)),\end{aligned}$$

where we used the fact (2.9) in the last equality. Hence, the assumption on λ guarantees that

$$\|f\|_{L_\Phi(A,\frac{m}{m(A)})} \le 2\lambda.$$

In other words,

$$\int_{\mathrm{supp}(f)} \Phi\left(\frac{|f(x)|}{2\lambda}\right)\frac{dx}{m(A)} \le 1.$$

Hence,

$$\int_{\mathbb{R}} \Phi\left(\frac{|f(x)|}{2\lambda}\right)dx \le m(A).$$

According to Construction 6.1, we have

$$\begin{aligned}m(\Sigma_\lambda) &\le 3\cdot 2^6\int_{\mathbb{R}} \Phi\left(\frac{|f(x)|}{\lambda}\right)dx \le 3\cdot 2^8\int_{\mathbb{R}} \Phi\left(\frac{|f(x)|}{2\lambda}\right)dx\\ &\le 3\cdot 2^8 m(A) \le 2^{10}m(A).\end{aligned}$$

This is the desired assertion. □

Lemma 6.5 *Under Construction 6.1, if* $\lambda \ge \mathscr{C}^{\frac{3}{2}}\mu(f)(m(\mathrm{supp}(f)))$, *then*

$$\lambda m(\Sigma_\lambda) \le 2^{12}m(\mathrm{supp}(f))\cdot(\mathscr{C}^{\frac{3}{2}}\mu(f))(m(\mathrm{supp}(f))).$$

Proof For brevity, write $A = \mathrm{supp}(f)$. According to Construction 6.1, we have

$$\lambda m(\Sigma_\lambda) = \sum_{Q\in\mathcal{K}_\lambda}\lambda m(Q) \le \sum_{Q\in\mathcal{K}_\lambda} m(Q)\|f\|_{L_\Phi(Q,\frac{m}{m(Q)})}.$$

By Lemma A.2, the function $\psi(t) = 2t\log^{\frac{1}{2}}(\frac{e}{t})$ is a concave majorant for the fundamental function of $L_\Phi(0,1)$. By [6], there exists a measure-preserving map from

Σ_λ to the interval $[0, m(\Sigma_\lambda)]$. Under this map, the disjoint dyadic cubes $\{Q\}_{Q\in\mathcal{K}_\lambda}$ becomes a collection of pairwise disjoint measurable subsets in $[0, m(\Sigma_\lambda)]$. Applying Theorem 4.1 and taking into account that composition with a measure-preserving map which does not change μ, we obtain

$$\begin{aligned}
&\sum_{Q\in\Sigma_\lambda} m(Q)\|f\|_{L_\Phi(Q,\frac{m}{m(Q)})}\\
&\quad\le \int_0^{m(\Sigma_\lambda)} \mu(s,f)\psi'\left(\frac{s}{m(\Sigma_\lambda)}\right)ds\\
&\quad= 2\int_0^{m(\Sigma_\lambda)} \mu(s,f)\left(\log^{\frac12}\left(\frac{e\cdot m(\Sigma_\lambda)}{s}\right)-\frac12\log^{-\frac12}\left(\frac{e\cdot m(\Sigma_\lambda)}{s}\right)\right)ds\\
&\quad\le 2\int_0^{m(\Sigma_\lambda)} \mu(s,f)\log^{\frac12}\left(\frac{e\cdot m(\Sigma_\lambda)}{s}\right)ds.
\end{aligned}$$

By Lemma 6.4, we have

$$\begin{aligned}
\lambda m(\Sigma_\lambda) &\le 2\int_0^{2^{10}m(A)} \mu(s,f)\log^{\frac12}\left(\frac{e\cdot 2^{10}m(A)}{s}\right)ds\\
&\le 2^{11}\int_0^{m(A)} \mu(s,f)(3+\log^{\frac12}\left(\frac{m(A)}{s}\right))ds\\
&= 2^{11}\Gamma(\frac{3}{2})m(A)\cdot(\mathscr{C}^{\frac32}\mu(f))(m(A))+2^{11}\cdot 3m(A)\cdot(\mathscr{C}\mu(f))(m(A))\\
&\le 2^{12}m(A)\cdot(\mathscr{C}^{\frac32}\mu(f))(m(A)).
\end{aligned}$$

This is exactly the required assertion. □

Next lemma shows that Marcinkiewicz multiplier T_{m} on $\mathbb{R}$ satisfies the property WCP($\frac32$).

Proposition 6.6 *Let T_{m} be a Marcinkiewicz multiplier on $\mathbb{R}$. Then T_{m} satisfies the property* WCP($\frac32$)*, that is, if $\lambda \ge \mathscr{C}^{\frac32}\mu(f)(m(\mathrm{supp}(f)))$, then*

$$\lambda m\left(\left\{y\in\mathbb{R}:\ |T_{\mathsf{m}}f|(y)>\lambda\right\}\right)\le c_{T_{\mathsf{m}}}\cdot m(\mathrm{supp}(f))\cdot(\mathscr{C}^{\frac32}\mu(f))(m(\mathrm{supp}(f)))$$

for each $0\le f\in L_\Phi(\mathbb{R})$ supported on a set of finite measure.

Proof The assertion immediately follows from Theorem 6.2, Lemma 6.5 and the fact (5.2). □

Now we are ready to prove Theorem 1.4.

Proof of Theorem 1.4 By Proposition 6.6 and Lemma 5.7, we have

$$\|T_{\mathsf{m}}\|_{L_p(\mathbb{R})\to L_p(\mathbb{R})}\le c_{T_{\mathsf{m}}}(p')^{\frac32},\quad 1<p\le 2.$$

By duality,

$$\|T_{\mathsf{m}}\|_{L_p(\mathbb{R})\to L_p(\mathbb{R})} \le c_{T_{\mathsf{m}}} p^{\frac{3}{2}}, \quad 2 \le p < \infty.$$

Combining the above estimates and Proposition 6.6, the desired assertion for T_{m} immediately follows from Theorem 3.1. □

6.2 Proof of Theorem 1.5

In this subsection, we show Theorem 1.5. Let $\{r_k\}_{k\ge1}$ be the Rademacher functions on $[0, 1]$ given by

$$r_k(t) = \operatorname{sgn}\sin(2^k\pi t), \quad t \in [0, 1].$$

It follows from [2, Chapter 1] that $\{r_k\}_{k\ge1}$ are independent random variables on $[0, 1]$ with probability $\frac{1}{2}$ for $r_k = 1$ and also probability $\frac{1}{2}$ for $r_k = -1$. The following assertion is Lemma 2.3 in [2].

Lemma 6.7 *For every real-valued sequence $\{a_k\}_{k\ge1} \in \ell_2$, we have*

$$\mu\left(\frac{1}{36}, \sum_{k\ge1} a_k r_k\right) \ge \frac{1}{\sqrt{2}}\left(\sum_{k\ge1} a_k^2\right)^{\frac{1}{2}}.$$

Consequently, we have the following result.

Lemma 6.8 *For every real-valued function sequence $\{f_k\}_{k\ge1} \in L_2(\Omega; \ell_2)$ with $\Omega \subset \mathbb{R}$, we have*

$$\mu^{\frac{1}{2}}\left(\sum_{k\ge1} f_k^2\right) \le \sqrt{2}\sigma_{36}\mu\left(\sum_{k\ge1} f_k \otimes r_k\right),$$

where function $f_k \otimes r_k$ is defined as $(f_k \otimes r_k)(x, t) = f_k(x)r_k(t)$ for $x \in \Omega$ and $t \in [0, 1]$.

Proof Since the distribution of a function equals the distribution of its decreasing rearrangement function, it follows that, for any $\lambda > 0$,

$$\begin{aligned} m_{\Omega\times[0,1]}\left(\left\{|\sum_{k\ge1} f_k \otimes r_k| > \lambda\right\}\right) &= m\left(\left\{(x,t) \in \Omega\times[0,1] : \left|\sum_{k\ge1} f_k(x)r_k(t)\right| > \lambda\right\}\right) \\ &= \int_\Omega m\left(\left\{t \in [0,1] : \left|\sum_{k\ge1} f_k(x)r_k(t)\right| > \lambda\right\}\right) dx \\ &= \int_\Omega m\left(\left\{t \in [0,1] : \mu\left(t, \sum_{k\ge1} f_k(x)r_k\right) > \lambda\right\}\right) dx, \end{aligned}$$

where $m_{\Omega\times[0,1]}$ stands for the Lebesgue measure on $\Omega \times [0, 1]$. Using Lemma 6.7, we estimate

$$m\left(\left\{t \in [0,1] : \mu\left(t, \sum_{k\geq 1} f_k(x) r_k\right) > \lambda\right\}\right)$$
$$\geq m\left(\left\{t \in [0,1] : \frac{1}{\sqrt{2}}\left(\sum_{k\geq 1} f_k^2(x)\right)^{\frac{1}{2}} \chi_{[0,\frac{1}{36}]}(t) > \lambda\right\}\right).$$

Thus,

$$m_{\Omega\times[0,1]}\left(\left\{|\sum_{k\geq 1} f_k \otimes r_k| > \lambda\right\}\right)$$
$$\geq \int_\Omega m\left(\left\{t \in [0,1] : \frac{1}{\sqrt{2}}\left(\sum_{k\geq 1} f_k^2(x)\right)^{\frac{1}{2}} \chi_{[0,\frac{1}{36}]}(t) > \lambda\right\}\right) dx$$
$$= \frac{1}{36} m\left(\left\{\frac{1}{\sqrt{2}}\left(\sum_{k\geq 1} f_k^2\right)^{\frac{1}{2}} > \lambda\right\}\right).$$

The desired assertion now follows from the definition of decreasing rearrangement function. □

We also need a noncommutative generalization of Theorem 3.1 (i). We prefer not to say "the proof is similar to that of Theorem 3.1 (i)" but instead derive the noncommutative result as a corollary of the commutative one.

Lemma 6.9 *Let $\alpha, \beta > 0$. Let $(\mathcal{M}, \tau)$ be an atomless semifinite von Neumann algebra with $\tau(1) = \infty$. Let $T : L_2(\mathbb{R}) \to L_2(\mathcal{M}, \tau)$ be a linear contraction. Suppose that T admits a bounded linear extension from $L_p(\mathbb{R})$ to $L_p(\mathcal{M}, \tau)$ for all $1 < p < \infty$. If*

$$\|T\|_{L_p(\mathbb{R})\to L_p(\mathcal{M},\tau)} \leq \begin{cases} (p')^\alpha, & 1 < p \leq 2 \\ p^\beta, & 2 \leq p < \infty \end{cases}, \tag{6.1}$$

then there exists a positive constant $c_{\alpha,\beta}$ such that

$$\mu(Tf) \prec\prec c_{\alpha,\beta}(\mathscr{C}^\alpha + (\mathscr{C}^*)^\beta)\mu(f), \quad f \in \Lambda_{\psi_{\alpha,\beta}}(\mathbb{R})$$

where $\mu(Tf)$ stands for the generalised singular value function of operator Tf with respect to von Neumann algebra $(\mathcal{M}, \tau)$ (see, e.g. [34], for more information).

Proof Fix $f \in \Lambda_{\psi_{\alpha,\beta}}(\mathbb{R})$ and consider a commutative von Neumann algebra $\mathcal{A} \subset \mathcal{M}$ isomorphic to $L_\infty(\mathbb{R})$ such that $\tau|_{\mathcal{A}}$ is semifinite and such that $|Tf|$ is affiliated to $\mathcal{A}$ (see e.g. [12, Lemma 2.1] for the existence). Fix a trace-preserving $*$-isomorphism $\iota : (\mathcal{A}, \tau) \to L_\infty(\mathbb{R})$. Let $\mathrm{Exp}_{\mathcal{A}} : (\mathcal{M}, \tau) \to (\mathcal{A}, \tau)$ be the conditional expectation. Let $Tf = U|Tf|$ be a polar decomposition and let $\lambda(U^*)$ be the left multiplication operator by U^*. Define an operator $T' : L_2(\mathbb{R}) \to L_2(\mathbb{R})$ by setting

$$T' = \iota \circ \mathrm{Exp}_{\mathcal{A}} \circ \lambda(U^*) \circ T.$$

Clearly,

$$\|T'\|_{L_p(\mathbb{R}) \to L_p(\mathbb{R})} \le \begin{cases} (p')^\alpha, & 1 < p \le 2 \\ p^\beta, & 2 \le p < \infty \end{cases}.$$

By Theorem 3.1 (i), we have

$$\mu(T'h) \prec\prec c_{\alpha,\beta}(\mathscr{C}^\alpha + (\mathscr{C}^*)^\beta)\mu(h)$$

for every $h \in \Lambda_{\psi_{\alpha,\beta}}(\mathbb{R})$. Taking into account that $\mu(T'f) = \mu(Tf)$ and using the last inequality with $h = f$, we complete the proof. □

Now we are ready to prove Theorem 1.5.

Proof of Theorem 1.5 Fix $n \in \mathbb{N}$. Given a sequence of signs $\{\epsilon_k\}_{k\in\mathbb{Z}}$, consider the Marcinkiewicz multipliers defined for $f \in \mathrm{dom}(\mathscr{C}^{\frac{3}{2}}) \cap \mathrm{dom}((\mathscr{C}^*)^{\frac{3}{2}})$ on $\mathbb{R}$ by

$$T_\epsilon f = \sum_{k\in\mathbb{Z}} \epsilon_k \chi_{L_k}(D) f, \quad T_{\epsilon,n} f = \sum_{k=1}^{n} \epsilon_k \chi_{L_k}(D) f,$$

where $L_k = (-2^{k+1}, -2^k] \cup [2^k, 2^{k+1})$ is the Littlewood–Paley interval, and $\chi_{L_k}(D)$ denotes the spectral measure of the differential operator D on $\mathbb{R}$. By Theorem 1.4, we have

$$\mu(T_{\epsilon,n} f) \le c_{\mathrm{abs}}(\mathscr{C}^{\frac{3}{2}} + (\mathscr{C}^*)^{\frac{3}{2}})\mu(f). \tag{6.2}$$

In order to estimate $\mu(Sf)$, let us define

$$I_k^{+1} = \{s \in [0,1] : r_k(s) = 1\}, \quad I_k^{-1} = \{s \in [0,1] : r_k(s) = -1\}, \quad 1 \le k \le n.$$

Write

$$[0,1] = \bigcup_{\substack{i_k \in \{+1,-1\} \\ 1 \le k \le n}} (I_1^{i_1} \cap \cdots \cap I_n^{i_n}),$$

where the union is a disjoint union. Denote $I_\epsilon = I_1^{i_1} \cap \cdots \cap I_n^{i_n}$ for $\epsilon = (i_1, \cdots, i_n) \in \{+1, -1\}^n$. Since $\{r_k\}_{k\geq 1}$ are independent random variables, it gives that

$$m(I_\epsilon) = m(I_1^{i_1} \cap \cdots \cap I_n^{i_n}) = \prod_{k=1}^{n} m(I_k^{i_k}) = \frac{1}{2^n}.$$

Therefore, for any $\lambda > 0$,

$$\begin{aligned} & m_{[0,1]\times\mathbb{R}}\left(\left\{|\sum_{k=1}^{n} r_k \otimes \chi_{L_k}(D)f| > \lambda\right\}\right) \\ &= \sum_{\epsilon\in\{+1,-1\}^n} m_{I_\epsilon\times\mathbb{R}}\left(\left\{|\sum_{k=1}^{n} \epsilon_k \chi_{L_k}(D)f| > \lambda\right\}\right) \\ &= \sum_{\epsilon\in\{+1,-1\}^n} m(I_\epsilon)\cdot m\Big(\{|T_{\epsilon,n}f| > \lambda\}\Big) \\ &= \frac{1}{2^n}\sum_{\epsilon\in\{+1,-1\}^n} m\Big(\{|T_{\epsilon,n}f| > \lambda\}\Big), \end{aligned}$$

where $m_{I\times\mathbb{R}}$ stands for the Lebesgue measure on $I \times \mathbb{R}$ with $I \subset \mathbb{R}$. Since the distribution of a function equals the distribution of its decreasing rearrangement function, we have

$$\begin{aligned} \frac{1}{2^n}\sum_{\epsilon\in\{+1,-1\}^n} m\Big(\{|T_{\epsilon,n}f| > \lambda\}\Big) &= \frac{1}{2^n}\sum_{\epsilon\in\{+1,-1\}^n} m\Big(\{\mu(T_{\epsilon,n}f) > \lambda\}\Big) \\ &= m_{[0,1]\times(0,\infty)}\left(\left\{\sum_{\epsilon\in\{+1,-1\}^n} \chi_{I_\epsilon}\otimes\mu(T_{\epsilon,n}f) > \lambda\right\}\right), \end{aligned}$$

where, in the second line, we use the fact that $\{I_\epsilon\}_{\epsilon\in\{+1,-1\}^n}$ are disjoint and their union is $[0, 1]$. Hence, by (6.2),

$$\begin{aligned} \mu\left(\sum_{k=1}^{n} r_k \otimes \chi_{L_k}(D)f\right) &= \mu\Big(\sum_{\epsilon\in\{+1,-1\}^n} \chi_{I\epsilon}\otimes\mu(T_{\epsilon,n}f)\Big) \\ &\leq c_{\mathrm{abs}}\mu\Big(\sum_{\epsilon\in\{+1,-1\}^n} \chi_{I\epsilon}\otimes(\mathscr{C}^{\frac{3}{2}} + (\mathscr{C}^*)^{\frac{3}{2}})\mu(f)\Big) \\ &= c_{\mathrm{abs}}(\mathscr{C}^{\frac{3}{2}} + (\mathscr{C}^*)^{\frac{3}{2}})\mu(f). \end{aligned}$$

Passing $n \to \infty$, the above estimate gives that

$$\mu\Big(\sum_{k\geq 1} r_k \otimes \chi_{L_k}(D)f\Big) \leq c_{\mathrm{abs}}(\mathscr{C}^{\frac{3}{2}} + (\mathscr{C}^*)^{\frac{3}{2}})\mu(f).$$

By Lemma 3.7 and 6.8, we have

$$\mu^{\frac{1}{2}}\left(\sum_{k\geq 1}|\chi_{L_k}(D)f|^2\right)\leq\sqrt{2}\sigma_{36}\mu\left(\sum_{k\in\mathbb{N}}r_k\otimes\chi_{L_k}(D)f\right)\leq c_{\rm abs}(\mathscr{C}^{\frac{3}{2}}+(\mathscr{C}^*)^{\frac{3}{2}})\mu(f).$$

When $k\leq 0$, taking r_{-k+1} and $T'_{\epsilon,n}f=\sum_{k=-n}^{0}\epsilon_k\chi_{L_k}(D)f$ into a consideration, and repeating the previous arguments, we obtain

$$\mu^{\frac{1}{2}}\left(\sum_{k\leq 0}|\chi_{L_k}(D)f|^2\right)\leq c_{\rm abs}(\mathscr{C}^{\frac{3}{2}}+(\mathscr{C}^*)^{\frac{3}{2}})\mu(f).$$

Therefore, according to the definition of S as in (1.4) and the last two inequalities, for each $t>0$, we have

$$\begin{aligned}\mu(2t,Sf)&\leq\mu^{\frac{1}{2}}\left(t,\sum_{k\leq 0}|\chi_{L_k}(D)f|^2\right)+\mu^{\frac{1}{2}}\left(t,\sum_{k\geq 1}|\chi_{L_k}(D)f|^2\right)\\&\leq c_{\rm abs}(\mathscr{C}^{\frac{3}{2}}+(\mathscr{C}^*)^{\frac{3}{2}})\mu(f)(t)\leq c_{\rm abs}(\mathscr{C}^{\frac{3}{2}}+(\mathscr{C}^*)^{\frac{3}{2}})\mu(f)(2t),\end{aligned}$$

where the last inequality is due to Lemma 3.7.

We also need some help from noncommutative analysis. Let $\mathcal{R}$ be a hyperfinite II_1 factor, and let $\{\omega_k\}_{k\in\mathbb{Z}}\subset\mathcal{R}$ be the sequence of Fermions, which satisfies

(i) Each ω_k is self-adjoint and unitary;
(ii) $\omega_k\omega_\ell=-\omega_\ell\omega_k, k\neq\ell$.

These properties can be found, for instance, in [29, p.458]. Consider an operator $T_{\mathcal{R}}:L_p(\mathbb{R})\to L_p(L_\infty(\mathbb{R})\bar{\otimes}\mathcal{R})$ defined by setting

$$T_{\mathcal{R}}:f\to\sum_{k\in\mathbb{N}}\omega_k\otimes\chi_{L_k}(D)f,\quad f\in{\rm dom}(\mathscr{C}^{\frac{3}{2}})\cap{\rm dom}((\mathscr{C}^*)^{\frac{3}{2}})$$

where the infinite summation is understood in the sense of $L_p(L_\infty(\mathbb{R})\bar{\otimes}\mathcal{R})$-norm $(1<p<\infty)$. It is clear that (denote by $1_{\mathcal{R}}$ the identity operator in $\mathcal{R}$)

$$|T_{\mathcal{R}}f|^2=(Sf)^2\otimes 1_{\mathcal{R}}.$$

We, therefore, have

$$\mu(T_{\mathcal{R}}f)=\mu(Sf)\leq c_{\rm abs}(\mathscr{C}^{\frac{3}{2}}+(\mathscr{C}^*)^{\frac{3}{2}})\mu(f).\tag{6.3}$$

Here, $\mu(T_{\mathcal{R}}f)$ is the generalised singular value function of operator $T_{\mathcal{R}}f$ associated to von Neumann algebra $L_\infty(\mathbb{R})\bar{\otimes}\mathcal{R}$ (see, e.g. [34], for more information).

Note that $T_{\mathcal{R}}$ is linear, and for $1\leq p\leq\infty$,

$$\|T_{\mathcal{R}}f\|_{L_p(L_\infty(\mathbb{R})\bar{\otimes}\mathcal{R})}=\|Sf\otimes 1_{\mathcal{R}}\|_{L_p(L_\infty(\mathbb{R})\bar{\otimes}\mathcal{R})}=\|Sf\|_{L_p(L_\infty(\mathbb{R}))}=\|Sf\|_{L_p(\mathbb{R})}.$$

Therefore, by Theorem D (Bourgain's result in [8]),

$$\|T_{\mathcal{R}}\|_{L_p(\mathbb{R})\to L_p(L_\infty(\mathbb{R})\bar{\otimes}\mathcal{R})} \le \begin{cases} p'^{\frac{3}{2}}, & 1 < p \le 2 \\ p, & 2 \le p < \infty \end{cases}.$$

Thus, by Lemma 6.9, we have

$$T_{\mathcal{R}} f \prec\prec c'_{\rm abs}(\mathscr{C}^{\frac{3}{2}} + \mathscr{C}^*)\mu(f). \tag{6.4}$$

Now, fix $t > 0$ and and decompose $f = f_1 + f_2$ such that

$$\mu(u, f_1) = \mu(u, f)\chi_{(0,t)}(u), \quad \mu(u, f_2) = \mu(u + t, f), \quad u > 0.$$

Obviously, $m(\mathrm{supp}(f_1)) = t$ and, therefore,

$$(\mathscr{C}^*)^{\frac{3}{2}}\mu(f_1)(t) = 0.$$

On the other hand, since $\mu(f_2) \le \min\{\mu(f), \mu(t, f)\}$, it follows that

$$\int_0^t \mathscr{C}^{\frac{3}{2}}\mu(f_2)(s)ds \le \mu(t, f)\frac{1}{\Gamma(\frac{3}{2})}\int_0^t \frac{1}{s}\int_0^s \left[\log\left(\frac{s}{u}\right)\right]^{1/2} duds = t\mu(t, f)$$

and

$$\int_0^t \mathscr{C}^*\mu(f_2)(s)ds \le t\mu(t, f) + t\mathscr{C}^*\mu(f)(t).$$

Combining these estimates and (6.3), (6.4), we have

$$\begin{aligned} \mu(2t, T_{\mathcal{R}} f) &\le \mu(t, T_{\mathcal{R}} f_1) + \mu(t, T_{\mathcal{R}} f_2) \\ &\le \mu(t, T_{\mathcal{R}} f_1) + \frac{1}{t}\int_0^t \mu(s, T_{\mathcal{R}} f_2)ds \\ &\le c_{\rm abs}((\mathscr{C}^{\frac{3}{2}} + (\mathscr{C}^*)^{\frac{3}{2}})\mu(f_1))(t) + c'_{\rm abs}\frac{1}{t}\int_0^t ((\mathscr{C}^{\frac{3}{2}} + \mathscr{C}^*)\mu(f_2))(s)ds \\ &\le (c_{\rm abs} + c'_{\rm abs})(\mathscr{C}^{\frac{3}{2}} + \mathscr{C}^*)\mu(f)(t). \end{aligned}$$

Now the expected upper estimate of $\mu(Sf)$ in Theorem 1.5 follows from the equality $\mu(Sf) = \mu(T_{\mathcal{R}} f)$ and Lemma 3.7. □

Remark 6.10 The above proof yields that the Marcinkiewicz multiplier T_ϵ has the following estimate:

$$\|T_\epsilon\|_{L_p(\mathbb{R})\to L_p(\mathbb{R})} \lesssim \frac{1}{(p-1)^{\frac{3}{2}}}, \quad 1 < p \le 2, \tag{6.5}$$

and

$$\|T_\epsilon\|_{L_p(\mathbb{R})\to L_p(\mathbb{R})} \lesssim p^{\frac{3}{2}}, \quad 2 < p < \infty. \tag{6.6}$$

However, Bourgain's result shows that $\|S\|_{L_p(\mathbb{R})\to L_p(\mathbb{R})} \sim p$ for $2 < p < \infty$. Hence, the distributional estimate of Sf is more delicate than the distributional estimate of $T_{\mathsf{m}} f$.

Appendix A: Additional estimates for Orlicz functions

In this appendix, we include the estimates for the fundamental functions of Orlicz spaces. These estimates are used in the proofs of Lemma 5.2 and 6.4 above.

Lemma A.1 *Let $\Phi(t) = t\log(1+t) + t$, $t > 0$, and let $\psi(t) = t\log(\frac{e}{t})$, $t \in (0,1)$. We have*

$$\frac{1}{\Phi^{-1}\left(\frac{1}{t}\right)} \le 2\psi(t), \quad t \in (0,1).$$

Proof Suppose first that $t \in (0, \frac{1}{6})$. Denoting $\frac{1}{t}$ by u, we observe that our inequality is equivalent to each of the following ones

$$\frac{1}{\Phi^{-1}(u)} \le \frac{2\log(eu)}{u} \Leftrightarrow \Phi^{-1}(u) \ge \frac{u}{2\log(eu)} \Leftrightarrow u \ge \Phi\left(\frac{u}{2\log(eu)}\right), \quad u > 6.$$

Note that

$$\Phi(s) = s\log(s) + s\log\left(1+\frac{1}{s}\right) + s \le s\log(s) + 1 + s \le s\left(\log(s)+2\right), \quad s > 1$$

and

$$\frac{u}{2\log(eu)} \ge \frac{6}{2\log(6e)} > 1, \quad u > 6.$$

Therefore, it suffices to prove

$$u \ge \frac{u}{2\log(eu)}\left(\log\left(\frac{u}{2\log(eu)}\right) + 2\right), \quad u > 6.$$

Dividing by u and multiplying by $2\log(eu)$, we rewrite the inequality as

$$2\log(eu) \ge \log\left(\frac{u}{2\log(eu)}\right) + 2 \Leftrightarrow \log(u)$$

$$\geq \log\left(\frac{1}{2\log(eu)}\right) \Leftrightarrow 2u\log(eu) \geq 1, \quad u > 6.$$

The latter inequality is obviously true. Hence, we establish the assertion for $t \in (0, \frac{1}{6})$.

Suppose now that $t \in [\frac{1}{6}, 1)$. We have

$$\frac{1}{\Phi^{-1}(\frac{1}{t})} \leq \frac{1}{\Phi^{-1}(1)} < \frac{2}{3}, \quad 2\psi(t) \geq 2\psi\left(\frac{1}{6}\right) = \frac{2}{6}\log(6e).$$

Taking into account that $\log(6e) > 2$, we establish the assertion for $t \in (\frac{1}{6}, 1)$. □

Lemma A.2 *Let $\Phi(t) = t\log^{\frac{1}{2}}(e+t)$, $t > 0$, and let $\psi(t) = t\log^{\frac{1}{2}}(\frac{e}{t})$, $t \in (0,1)$. We have*

$$\frac{1}{\Phi^{-1}(\frac{1}{t})} \leq 2\psi(t), \quad t \in (0,1).$$

Proof Suppose first $t \in (0, \frac{1}{2})$. Denoting $\frac{1}{t}$ by u, we observe that our inequality is equivalent to each of the following ones

$$\frac{1}{\Phi^{-1}(u)} \leq \frac{2\log^{\frac{1}{2}}(eu)}{u} \Leftrightarrow \Phi^{-1}(u) \geq \frac{u}{2\log^{\frac{1}{2}}(eu)} \Leftrightarrow u \geq \Phi\left(\frac{u}{2\log^{\frac{1}{2}}(eu)}\right), \quad u > 2.$$

In other words, we need to show

$$u \geq \frac{u}{2\log^{\frac{1}{2}}(eu)}\log^{\frac{1}{2}}\left(e + \frac{u}{2\log^{\frac{1}{2}}(eu)}\right) \Leftrightarrow 2\log^{\frac{1}{2}}(eu)$$
$$\geq \log^{\frac{1}{2}}\left(e + \frac{u}{2\log^{\frac{1}{2}}(eu)}\right), \quad u > 2.$$

It is immediate that

$$\log^{\frac{1}{2}}\left(e + \frac{u}{2\log^{\frac{1}{2}}(eu)}\right) \leq \log^{\frac{1}{2}}\left(e+u\right) \leq \log^{\frac{1}{2}}(eu), \quad u > 2.$$

Hence, we establish the assertion for $t \in (0, \frac{1}{2})$.

Suppose now that $t \in (\frac{1}{2}, 1)$. We have

$$\frac{1}{\Phi^{-1}(\frac{1}{t})} \leq \frac{1}{\Phi^{-1}(1)} < 1, \quad 2\psi(t) \geq 2\psi\left(\frac{1}{2}\right) = \log^{\frac{1}{2}}(2e) > 1.$$

This yields the assertion for $t \in (\frac{1}{2}, 1)$. □

Lemma A.3 *We have*

$$t\log^{\frac{1}{2}}\left(\frac{e}{t}\right) \le \frac{2}{\sqrt{\pi}}\int_0^t \log^{\frac{1}{2}}\left(\frac{1}{s}\right)ds, \quad 0 < t < 1.$$

Proof Substituting $t = e^{-u}$ and changing the variable $s = e^{-v}$ in the integral, we equivalently rewrite the assertion as

$$(u+1)^{\frac{1}{2}}e^{-u} \le \frac{2}{\sqrt{\pi}}\int_u^\infty v^{\frac{1}{2}}e^{-v}dv, \quad u > 0.$$

Set

$$f(u) = (u+1)^{\frac{1}{2}}e^{-u} - \frac{2}{\sqrt{\pi}}\int_u^\infty v^{\frac{1}{2}}e^{-v}dv, \quad u \ge 0.$$

We have $f(0) = f(\infty) = 0$ and

$$e^u f'(u) = \frac{1}{2}(u+1)^{-\frac{1}{2}} - (u+1)^{\frac{1}{2}} + \frac{2}{\sqrt{\pi}}u^{\frac{1}{2}}, \quad u \ge 0.$$

It is easy to see that f' has unique zero $u_0 \approx 0.58$ and $f'(0) < 0$. Hence, f decreases on $(0, u_0)$ and increases on (u_0, ∞). This completes the proof. □

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Declarations

Conflict of interest All authors declare that we have no conflict of interest.

References

1. Astashkin, S., Huang, J., Pliev, M., Sukochev, F., Zanin, D.: Optimal range of Haar martingale transforms and its applications. Int. Math. Res. Not. IMRN **2023**(24), 20827–20854 (2023)
2. Astashkin, S.V.: Rademacher functions in symmetric spaces. Sovrem. Mat. Fundam. Napravl. **32**, 3–161 (2009)
3. Bakas, O., Ciccone, V., Parissis, I., Vitturi, M.: Endpoint estimates for higher order Marcinkiewicz multipliers. Rev. Mat. Iberoam. (2024) **(online)**
4. Bennett, C., Rudnick, K.: On Lorentz-Zygmund spaces. Dissertationes Math. (Rozprawy Mat.) **175**, 67 (1980)
5. Bennett, C., Sharpley, R.: Weak-type inequalities for H^p and BMO, Harmonic analysis in Euclidean spaces (Proc. Sympos. Pure Math., Williams Coll., Williamstown, Mass., 1978), Part 1, Proc. Sympos. Pure Math., XXXV, Part 1, pp. 201–229. Amer. Math. Soc., Providence (1979)

6. Bennett, C., Sharpley, R.: Interpolation of Operators, Pure and Applied Mathematics, vol. 129. Academic Press Inc, Boston (1988)
7. Bergh, J., Löfström, J.: Interpolation Spaces. An Introduction, Grundlehren der Mathematischen Wissenschaften, No. 223. Springer, Berlin-New York (1976)
8. Bourgain, J.: On the Behavior of the Constant in the Littlewood–Paley Inequality, Geometric Aspects of Functional Analysis (1987–88), Lecture Notes in Math., vol. 1376, pp. 202–208. Springer, Berlin (1989)
9. Boyd, D.W.: The Hilbert transform on rearrangement-invariant spaces. Can. J. Math. **19**, 599–616 (1967)
10. Calderón, A.-P.: Spaces between L^1 and L^∞ and the theorem of Marcinkiewicz. Stud. Math. **26**, 273–299 (1966)
11. Calderon, A.P., Zygmund, A.: On the existence of certain singular integrals. Acta Math. **88**, 85–139 (1952)
12. Chilin, V., Sukochev, F.: Weak convergence in non-commutative symmetric spaces. J. Oper. Theory 35–65 (1994)
13. Christ, M.: Lectures on singular integral operators. In: CBMS Regional Conference Series. Mathematics, vol. 77, Published for the Conference Board of the Mathematical Sciences, Washington, DC; by the American Mathematical Society, Providence (1990)
14. Cianchi, A., Kerman, R., Opic, B., Pick, L.: A sharp rearrangement inequality for the fractional maximal operator. Stud. Math. **138**(3), 277–284 (2000)
15. Coifman, R.R., Rochberg, R., Weiss, G.: Factorization theorems for Hardy spaces in several variables. Ann. Math. (2) **103**(3), 611–635 (1976)
16. Cruz-Uribe, D., Martell, J., Pérez, C.: Weights, Extrapolation and the Theory of Rubio de Francia, Operator Theory: Advances and Applications, vol. 215. Birkhäuser/Springer Basel AG, Basel (2011)
17. Duong, X., Gong, R., Kuffner, M., Li, J., Wick, B., Yang, D.: Two weight commutators on spaces of homogeneous type and applications. J. Geom. Anal. **31**(1), 980–1038 (2021)
18. Edmunds, D.E., Mihula, Z., Musil, V., Pick, L.: Boundedness of classical operators on rearrangement-invariant spaces. J. Funct. Anal. **278**(4), 108341 (2020)
19. Edwards, R.E., Gaudry, G.I.: Littlewood–Paley and Multiplier Theory, Ergebnisse der Mathematik und ihrer Grenzgebiete [Results in Mathematics and Related Areas], vol. 90. Springer, Berlin-New York (1977)
20. Grafakos, L.: Modern Fourier Analysis, Graduate Texts in Mathematics, vol. 250, 2nd edn. Springer, New York (2009)
21. Hardy, G.H., Littlewood, J.E.: A maximal theorem with function-theoretic applications. Acta Math. **54**(1), 81–116 (1930)
22. Hardy, G.H., Littlewood, J.E., Pólya, G.: Inequalities, Cambridge Mathematical Library, Cambridge University Press, Cambridge, Reprint of the 1952 edition (1988)
23. Herz, C.: The Hardy–Littlewood maximal theorem. In: Symposium on Harmonic Analysis, University of Warwick, pp. 1–27 (1968)
24. Holmes, I., Lacey, M., Wick, B.: Commutators in the two-weight setting. Math. Ann. **367**, 51–80 (2017)
25. Holmstedt, T.: Interpolation of quasi-normed spaces. Math. Scand. **26**, 177–199 (1970)
26. Hytönen, T., van Neerven, J., Veraar, M., Weis, L.: Analysis in Banach spaces. Vol. I. Martingales and Littlewood–Paley theory, Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge. A Series of Modern Surveys in Mathematics [Results in Mathematics and Related Areas. 3rd Series. A Series of Modern Surveys in Mathematics], vol. 63. Springer, Cham (2016)
27. Jiao, Y., Quan, X., Sukochev, F., Zanin, D.: On noncommutative distributional Khintchine type inequalities. Bull. Lond. Math. Soc. **56**(7), 2278–2295 (2024)
28. Jiao, Y., Sukochev, F., Wu, L., Zanin, D.: Distributional inequalities for noncommutative martingales. J. Funct. Anal. **284**(5), 109798 (2023)
29. Jiao, Y., Sukochev, F., Zanin, D.: On subspaces spanned by freely independent random variables in noncommutative L_p-spaces. Isr. J. Math. **238**(1), 431–477 (2020)
30. Kalton, N.J.: Lattice structures on Banach spaces. Mem. Am. Math. Soc. **103**(493), vi+92 (1993)
31. Krein, S.G., Petunin, Y.I., Semenov, E.M.: Interpolation of Linear Operators, Translations of Mathematical Monographs, vol. 54. American Mathematical Society, Providence (1982)

32. Lindenstrauss, J., Tzafriri, L.: Classical Banach spaces. II, Ergebnisse der Mathematik und ihrer Grenzgebiete [Results in Mathematics and Related Areas], Function spaces, vol. 97. Springer, Berlin-New York (1979)
33. Long, R.: Martingale Spaces and Inequalities. Peking University Press, Beijing, Friedr. Vieweg & Sohn, Braunschweig (1993)
34. Lord, S., Sukochev, F., Zanin, D.: Singular Traces: Theory and Applications, vol. 46. Walter de Gruyter (2012)
35. O'Neil, R.: Convolution operators and $L(p,\ q)$ spaces. Duke Math. J. **30**, 129–142 (1963)
36. O'Neil, R., Weiss, G.: The Hilbert transform and rearrangement of functions. Stud. Math. **23**, 189–198 (1963)
37. Pérez, C.: Endpoint estimates for commutators of singular integral operators. J. Funct. Anal. **128**(1), 163–185 (1995)
38. Pick, L., Kufner, A., John, O., Fucík, S.: Function spaces, vol. 1. Walter de Gruyter (2012)
39. Riesz, F.: Sur un Theoreme de Maximum de Mm. Hardy et Littlewood. J. Lond. Math. Soc. **7**(1), 10–13 (1932)
40. Semënov, E.M., Shtekert, B.: Permutations of the haar system in spaces L_p. Anal. Math. **7**(4), 277–295 (1981)
41. Sharpley, R.: Counterexamples for classical operators on Lorentz–Zygmund spaces. Stud. Math. **68**(2), 141–158 (1980)
42. Stein, E.M.: The development of square functions in the work of A. Zygmund. Bull. Am. Math. Soc. **7**(2), 359–376 (1982)
43. Sukochev, F., Tulenov, K., Zanin, D.: The optimal range of the Calderòn operator and its applications. J. Funct. Anal. **277**(10), 3513–3559 (2019)
44. Tao, T., Wright, J.: Endpoint multiplier theorems of Marcinkiewicz type. Rev. Mat. Iberoam. **17**(3), 521–558 (2001)
45. Weisz, F.: Martingale Hardy Spaces and Their Applications in Fourier Analysis, Lecture Notes in Mathematics, vol. 1568. Springer, Berlin (1994)
46. Yano, S.: Notes on Fourier analysis. XXIX. An extrapolation theorem. J. Math. Soc. Japan **3**, 296–305 (1951)

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